

Topology-Preserving Lossy Compression

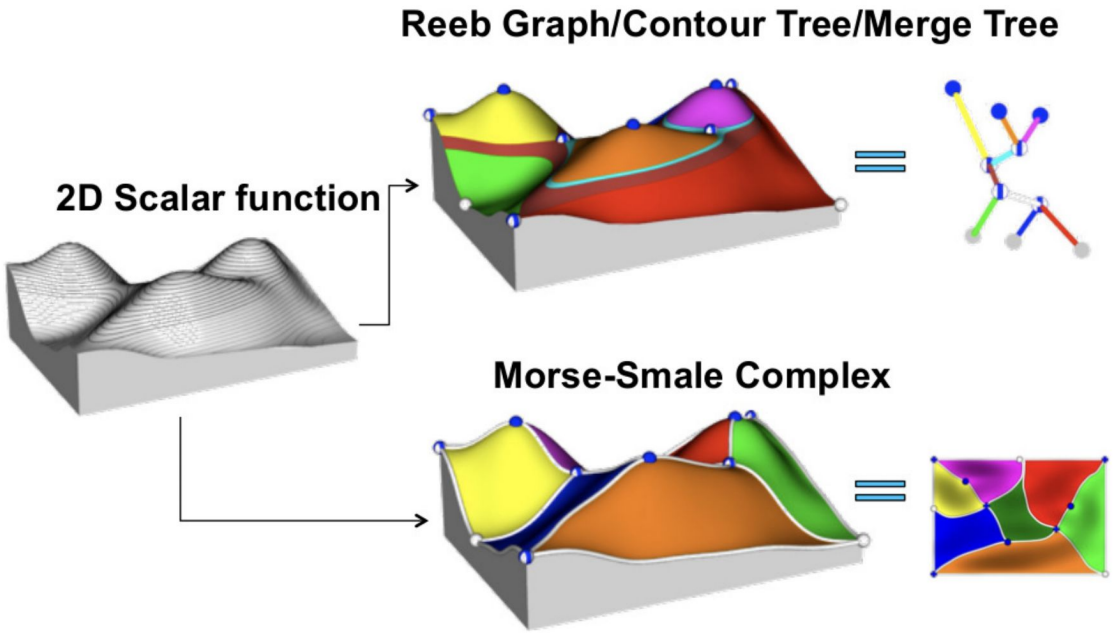
IEEE VIS 2026 Tutorial: Lossy Compression for Scientific Data

What this part covers

1	**Basic Concepts**	A crash course on topology: Critical points, merge and contour trees, extremum graphs, Morse–Smale segmentations and complexes, persistence diagrams
2	**Motivation**	Why a pointwise error bound does not protect any of these
3	**Scalar fields**	TopoQZ, TopoSZ, TopoA, MSz, pMSz, DMTz, EXaCTz
4	**Vector and tensor fields**	cpSZ, cpSZ-SoS, TspSZ, critical point trajectories, TFZ
5	**Summary**	Which method for which need, and open problems

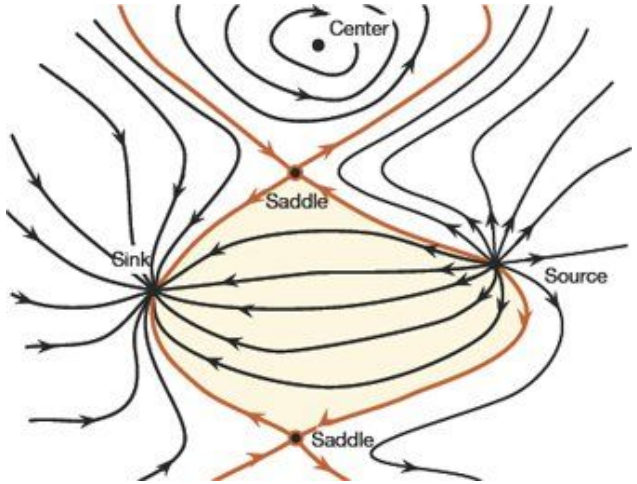
A Crash Course on Topological Feature Descriptors

Scalar Field Topology



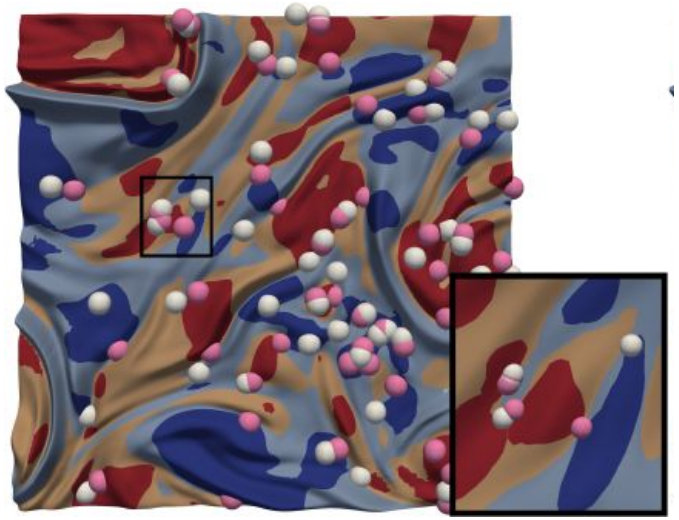
Topological feature descriptors for scalar field visualization (Liu et al. 2017)

Vector Field Topology



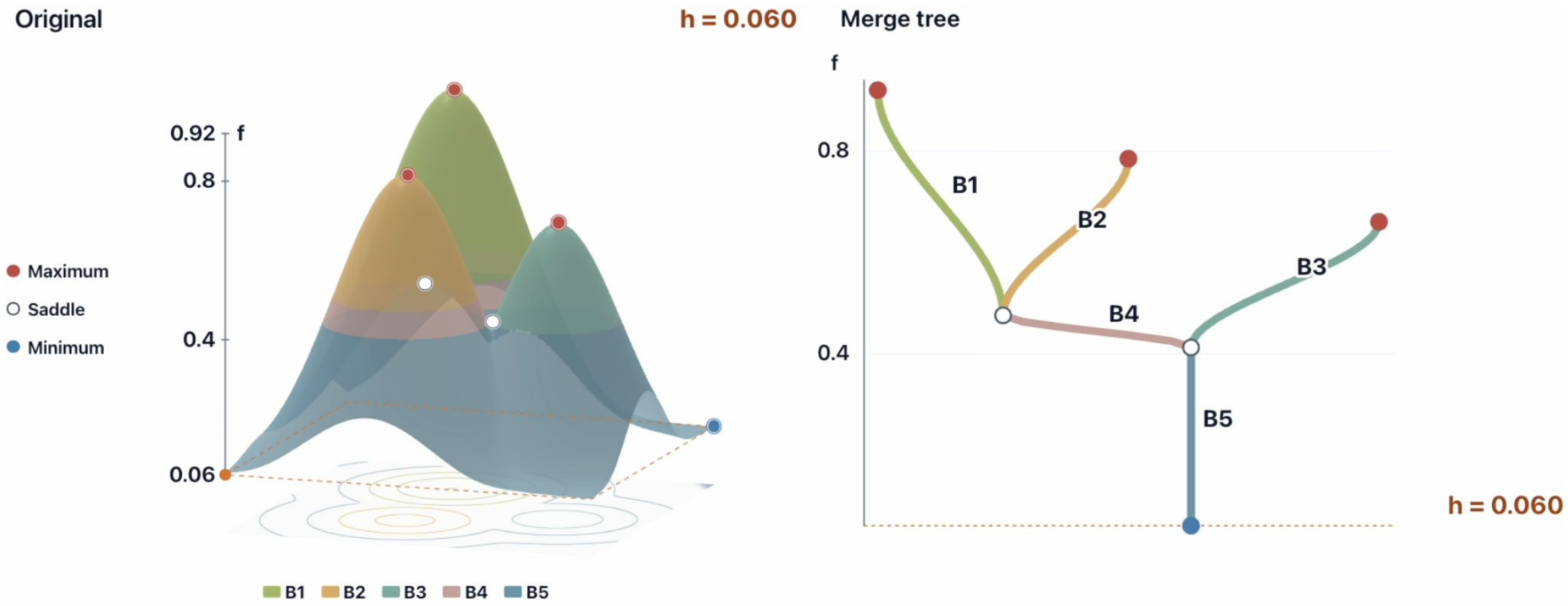
State of the Art in Time-Dependent Flow Topology: Interpreting Physical Meaningfulness Through Mathematical Properties (Bujack et al.)

Tensor Field Topology

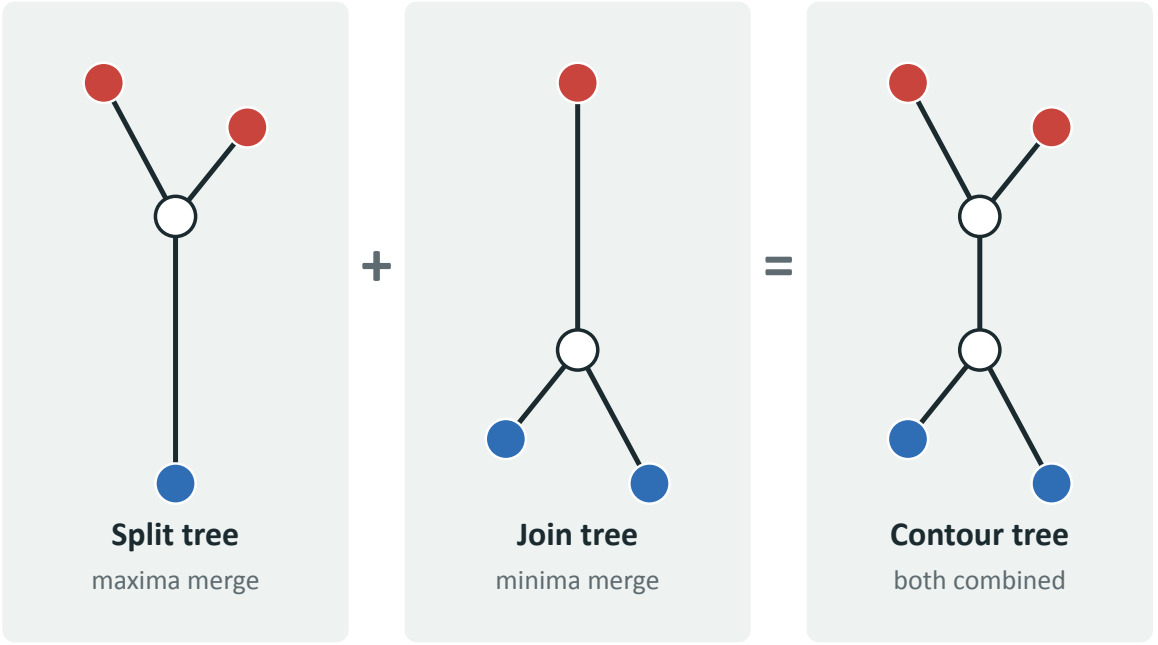


TFZ: Topology-Preserving Compression of 2D Symmetric and Asymmetric Second-Order Tensor Fields (Gorski et al.)

Merge Trees

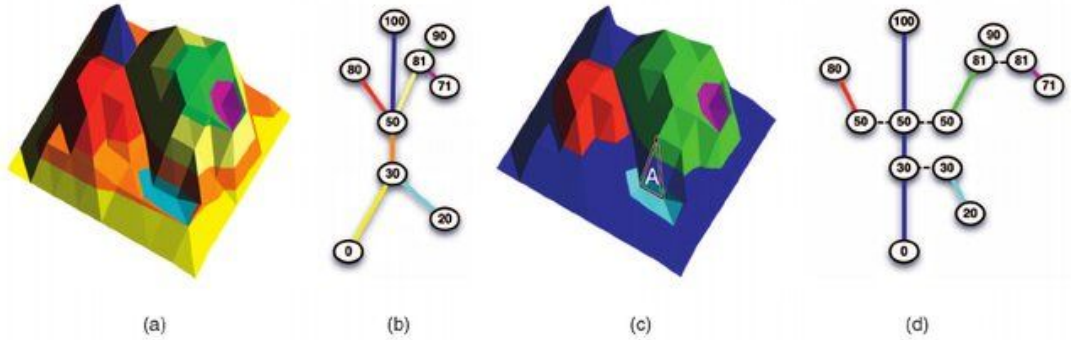

Original
h = 0.060
Merge tree
Maximum
Saddle
Minimum
0.92
f
0.8
0.4
0.06
B1
B2
B3
B4
B5
f
0.8
0.4
B1
B2
B3
B4
B5
h = 0.060

Contour Trees



What it is used for

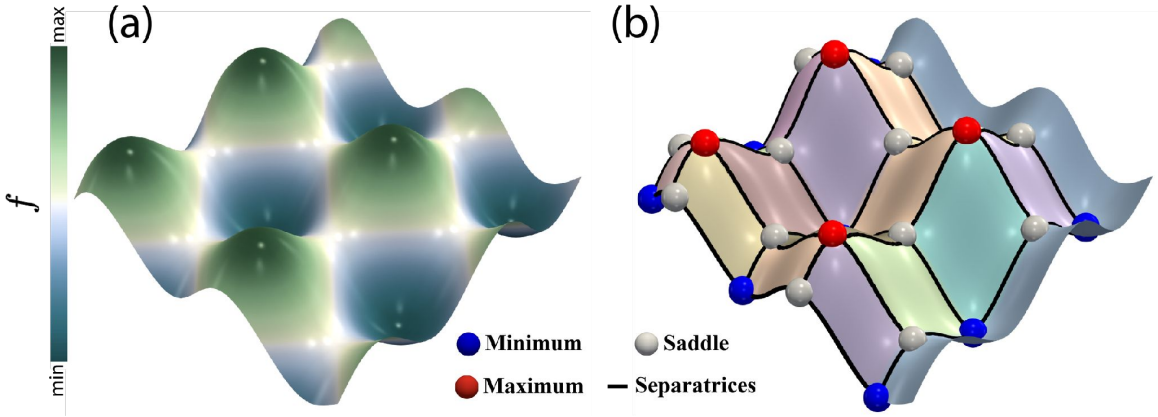
- A compact summary of all features and how they nest



Carr, Snoeyink and Axen, Computing contour trees in all dimensions, Computational Geometry 2003
Weber et al., Topology-Controlled Volume Rendering, IEEE TVCG, 13(2):330-341, 2007

Morse-Smale Complexes

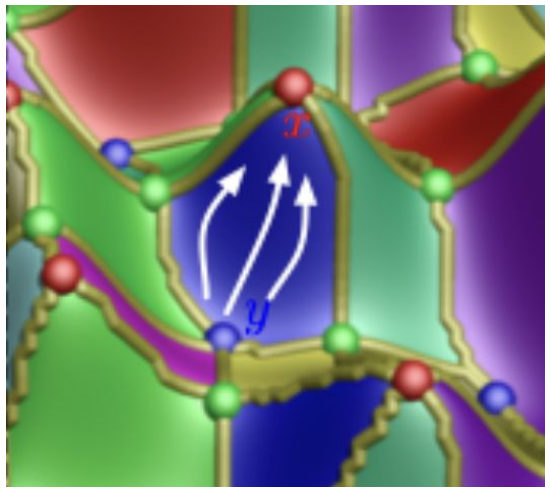
- Descending manifold of a maximum: everything that flows up to it
- Ascending manifold of a minimum: everything that flows down to it
- Their intersections are the Morse–Smale cells
- Separatrices: the integral lines from saddles to extrema that bound the cells



Topology-based Quality Metrics

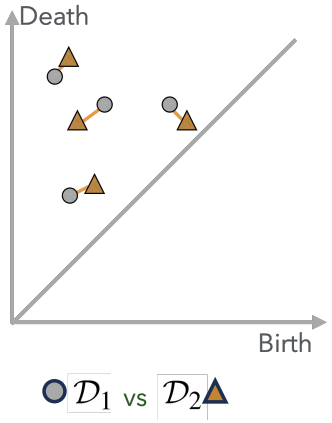
- Compression ratio
- PSNR, MSE, SSIM, etc.
- Runtime

- Topological descriptors
  e.g., Morse-Smale Complex (MSC)



  - Integral lines
  - Morse-Smale cells

- Distance between original f & decompressed f’



  - Bottleneck distance

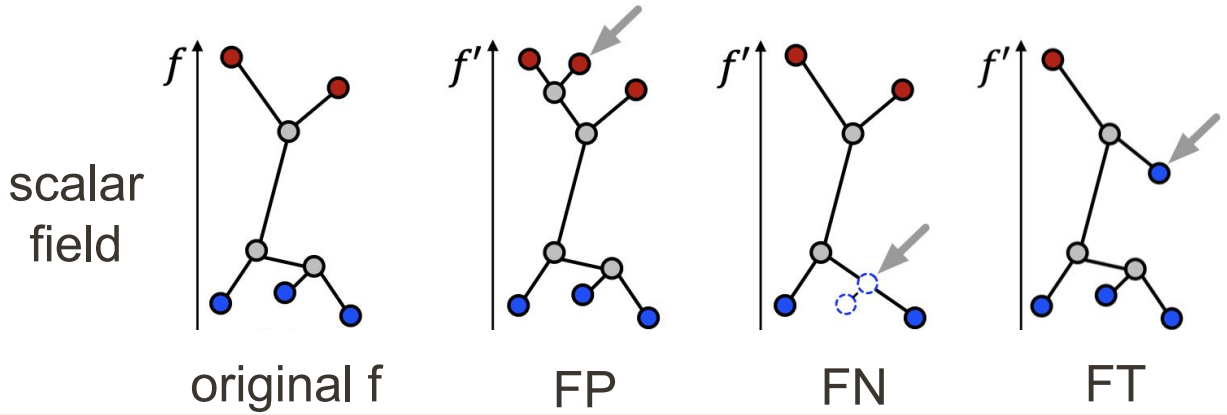
$$d_\infty(\mathcal{D}_1,\mathcal{D}_2) = \inf_{\eta:\mathcal{D}_1\to\mathcal{D}_2} \sup_{x\in\mathcal{D}_1} ||x-\eta(x)||_\infty.$$

  - p-Wasserstein distance

$$d_p(\mathcal{D}_1,\mathcal{D}_2) = \left[\inf_{\eta:\mathcal{D}_1\to\mathcal{D}_2} \sum_{x\in\mathcal{D}_1} ||x-\eta(x)||_\infty^p\right]^{\frac{1}{p}}$$

[Soler et al., PacificVis 2018]

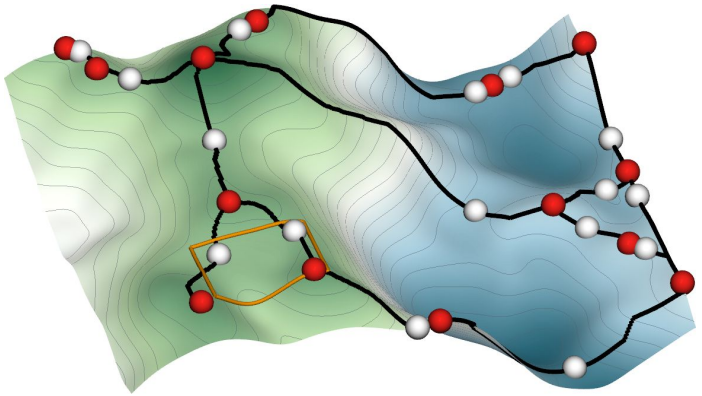
- False Positive (FP), False Negative (FN), False Type (FT)



[Yan et al., VIS 2023]

Compression Distorts Topology

Original Data · 256 GB

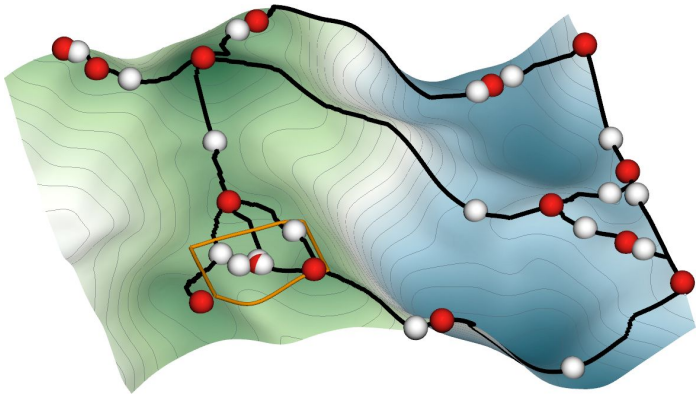


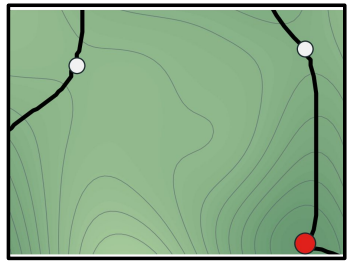
Compress with SZ3

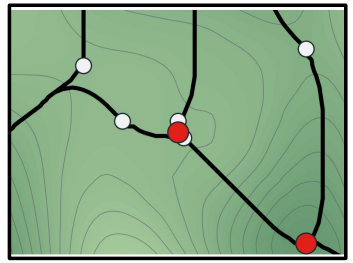
relative error bound 10^{-3}

≈ 12 GB compressed (≈ 21× smaller)

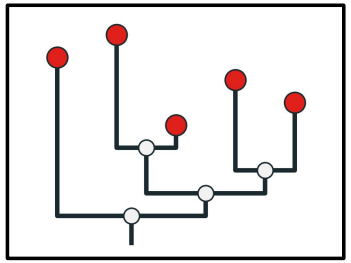
Decompressed Data

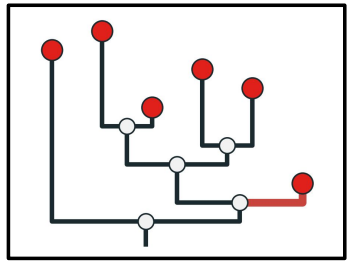






orig. ⟶ decomp.: **a false maximum appears (extremum graph)**

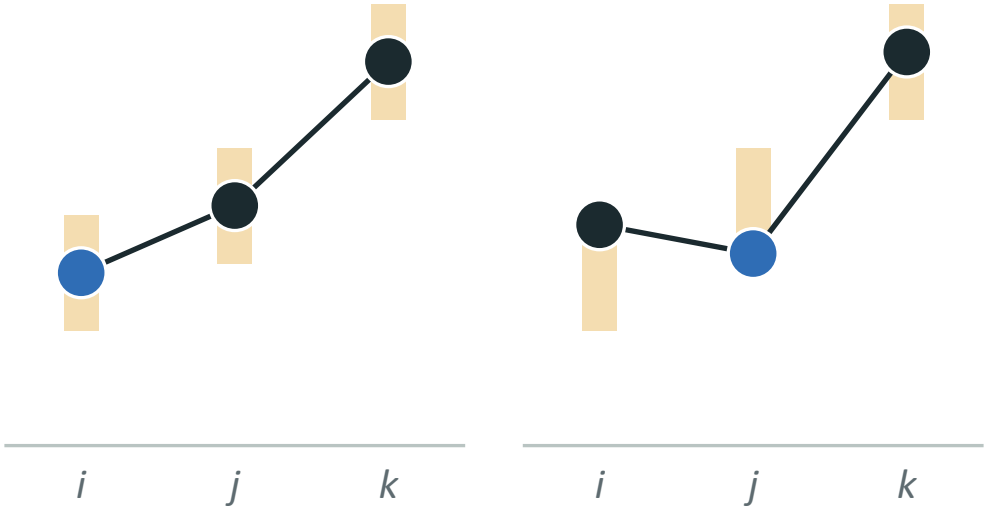




orig. ⟶ decomp.: **a new branch appears (merge tree)**

Ratio: SZ3 on Nyx at 10^{-3} (Liu et al., arXiv:2310.14133). Slice: illustrative field, SZ3 at 10^{-3}; tree heights by order.

Bounding Pointwise Error won't Preserve Topology



Original

i is the minimum

Decompressed

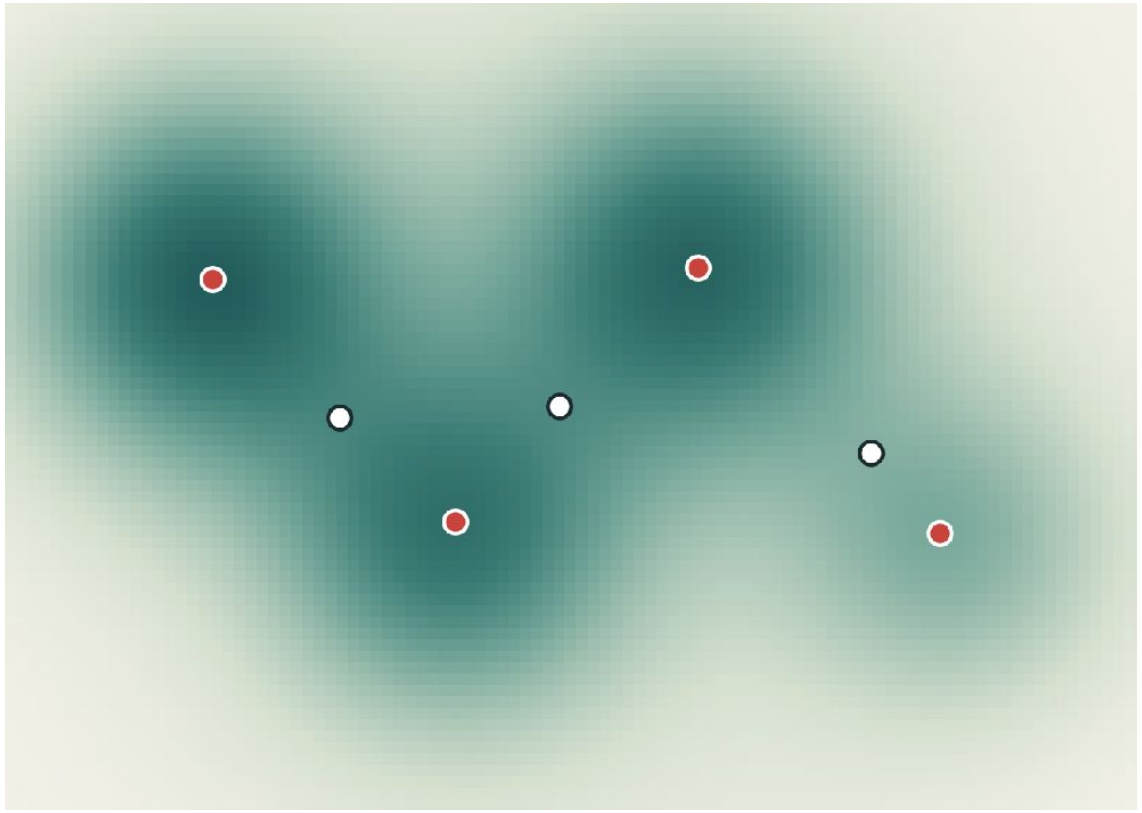
the minimum moved to j

Three types of false cases

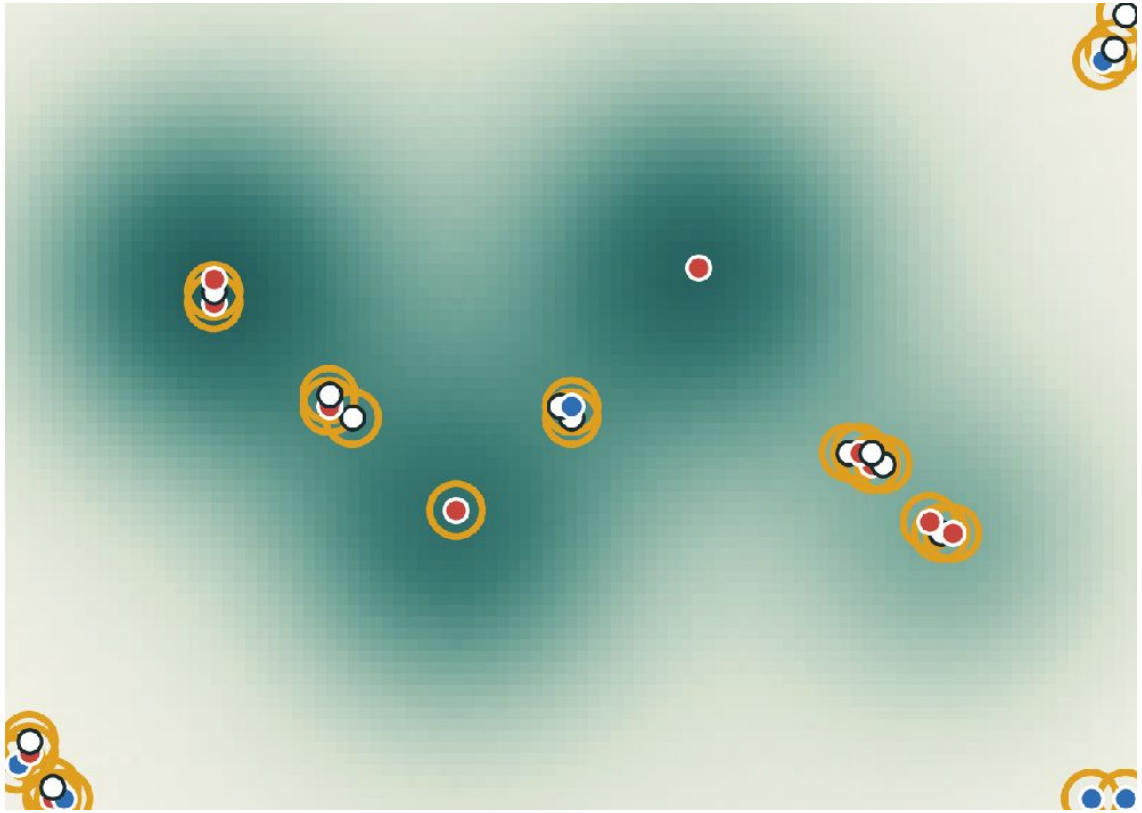
False positive	a critical point that is not in the original (j above)
False negative	an original critical point that disappears (i above)
False type	a critical point that changes type, for example a saddle that becomes a maximum

Both values stayed within ±ξ. Their order flipped, and the topology changed.

Bounding Pointwise Error won't Preserve Topology



Original: 7 critical points



Errors ≤ 0.5% of the range: 30 critical points

Synthetic illustration. Rings mark critical points that differ from the original.

State-of-the-Art in Topology Preserving Compression

	Descriptors	Methods
Scalar fields	**Merge trees / contour trees**	TopoQZ [Soler 2018], TopoSZ [Yan 2024], TopoA [Gorski 2025]
	Morse–Smale segmentation	MSz [Li 2025], pMSz [Li, 2026]
	Morse–Smale complexes	DMTz [Li 2026]
	Extremum graphs	EXaCTz [Li, 26]
Vector fields	**Critical points (and trajectories)**	[Liang 2020], [Xia 2024]
	Topological skeleton	TspSZ [Xia, 25]
Tensor Fields	**Degeneracies**	TFZ [Gorski 2025]

A Taxonomy for Topology-Preserving Compression

Modify the compressor

Derive a per-point error bound (or quantization) from the topology, so that the compressor cannot break it.

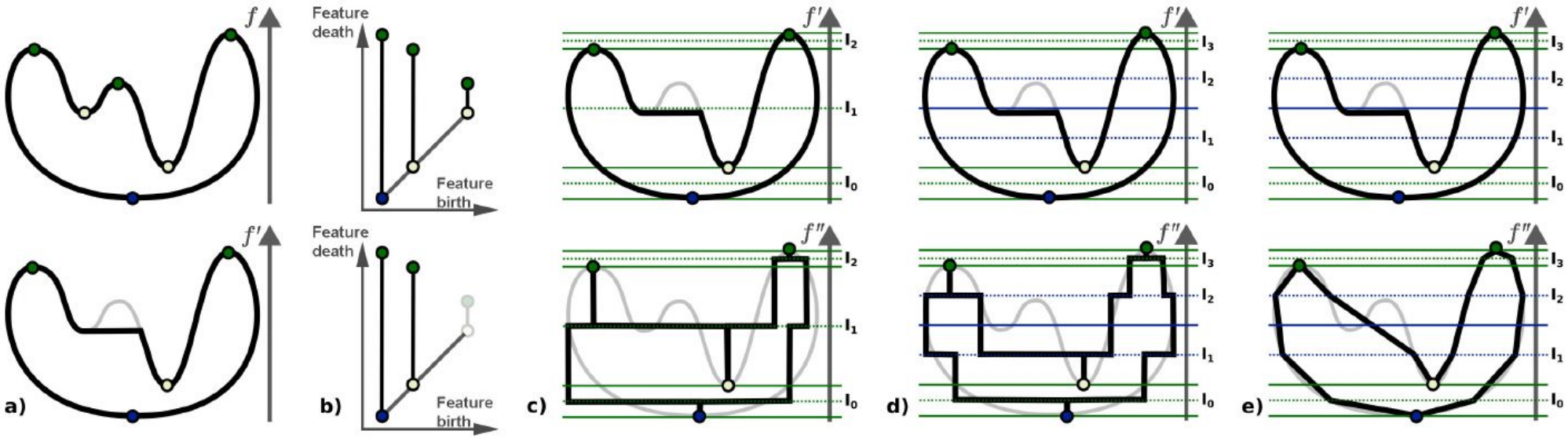
TopoQZ, TopoSZ, cpSZ, cpSZ-SoS, TspSZ, TFZ

Correct the decompressed data

Take the output of any error-bounded compressor and fix the topology within the bound.

TopoA, MSz, pMSz, DMTz, EXaCTz

TopoQZ: Compression Controlled by Persistence

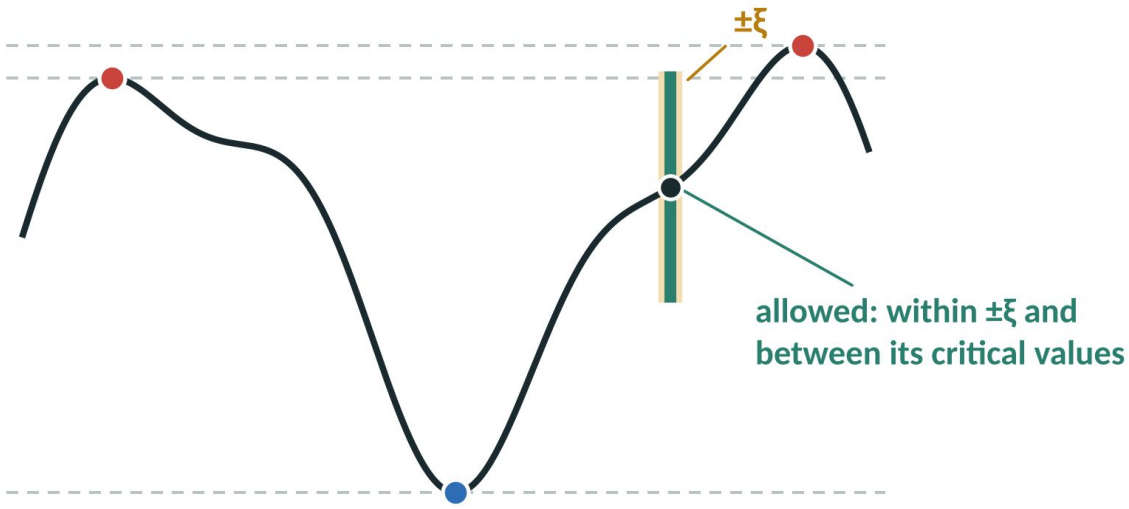


- Methodology: Pre-simplify by removing critical-point pairs with persistence below ε before quantization
- Pairs above ε kept exactly so the bottleneck distance ≤ ε

Figure and method: Soler, Plainchault, Conche and Tierny, Topologically controlled lossy compression, IEEE PacificVis 2018

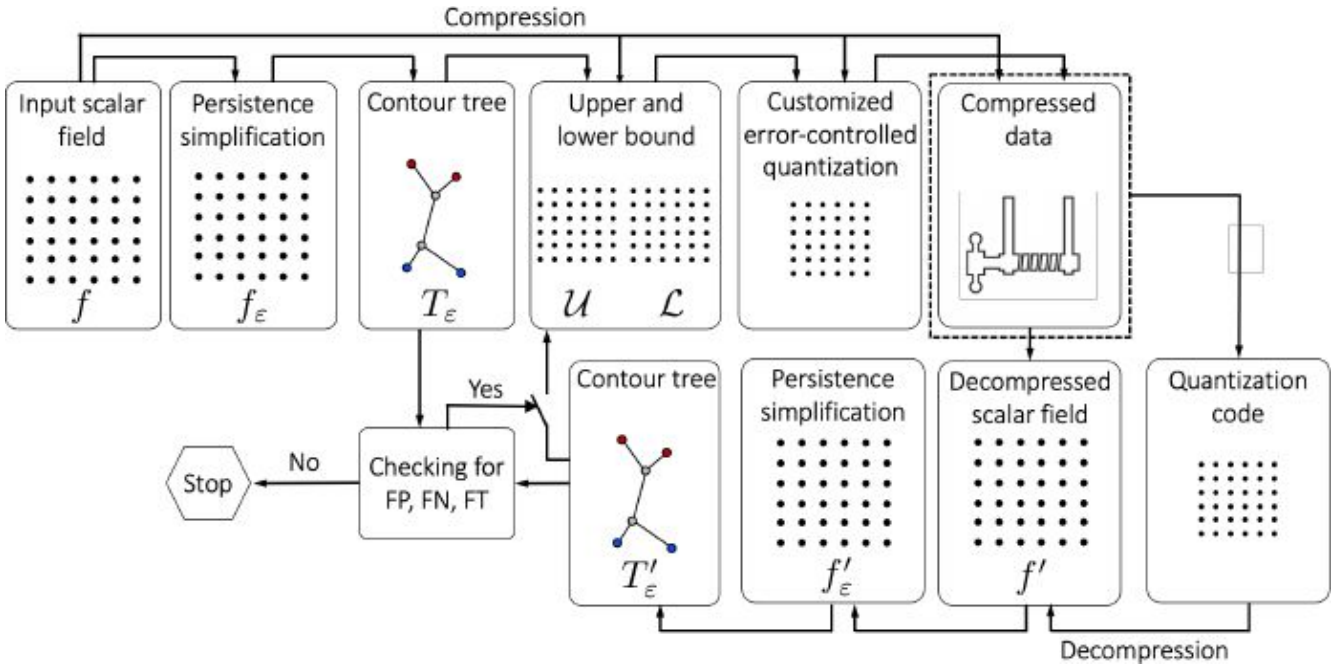
TopoSZ: Adding topological constraints into the SZ compressor

- Each vertex gets a lower and upper bound from its contour tree segment
- It can move within ±ξ, but never past a critical value: the contour tree cannot change



Yan, Liang, Guo and Wang, TopoSZ: Preserving topology in error-bounded lossy compression, IEEE TVCG 2024 (VIS 2023)

TopoSZ: Adding topological constraints into the SZ compressor



Deriving localized error bounds

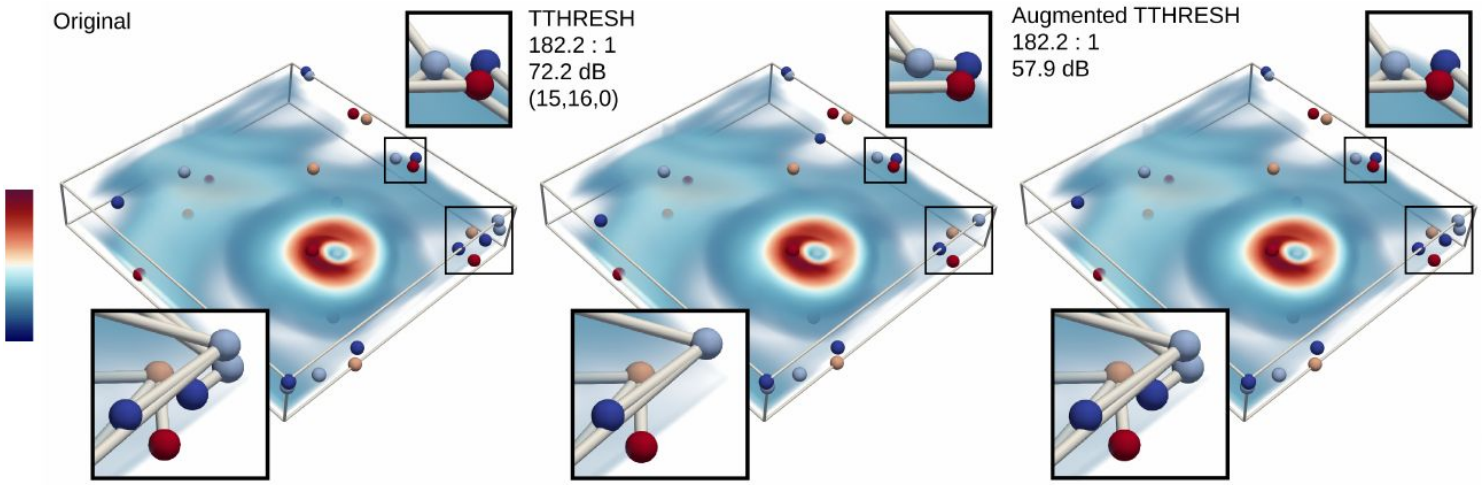
Determining lower and upper error bounds for every single value based on saddle-extrema pairs

Checking topology consistency iterative

Identifying new inconsistencies, tightening local error bounds, and verify again

Figure: Yan, Liang, Guo and Wang, TopoSZ, IEEE TVCG 2024

TopoA: contour tree guarantees for any compressor



Augmenting any compressor

A more generic framework that can augment any lossy compressors, e.g., SZ3, ZFP, TTHRESH, neural compressors

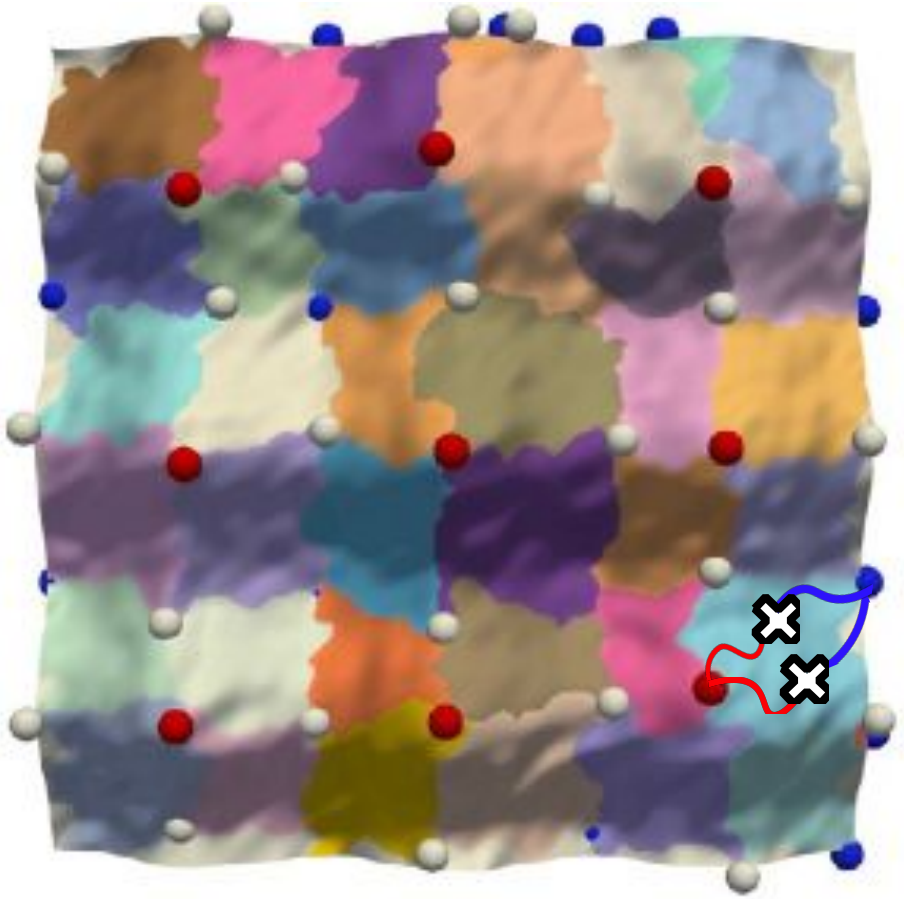
Tightening local error bounds

Iteratively tightening local error bounds and verifying against the ground truth until convergence

Figure: Gorski, Liang, Guo, Yan and Wang, A general framework for augmenting lossy compressors with topological guarantees, IEEE TVCG 2025. Timing: Li et al., EXaCTz.

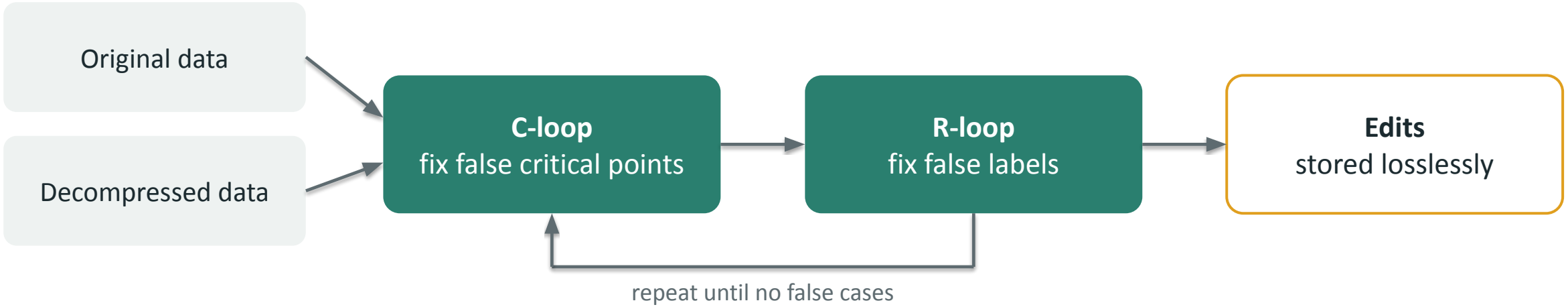
MSz: Preserving Morse-Smale Segmentations

- Morse-Smale segmentations divide the domain into monotonous segmentations, where the data within each segmentation is monotonous. Key components are:
    - **Extrema**: Maxima / Minima
    - **Ascending/descending** manifolds
- Vertices that flows to the same min/max are labeled for the same segmentation
- Preservation strategy
    - Fix wrong extrema and labels by editing values
    - Edits stay within ξ, stored with the data



Li, Liang, Wang, Qiu, Yan and Guo, MSz, IEEE TVCG 2025 (VIS 2024)

MSz workflow: C-loops and R-loops



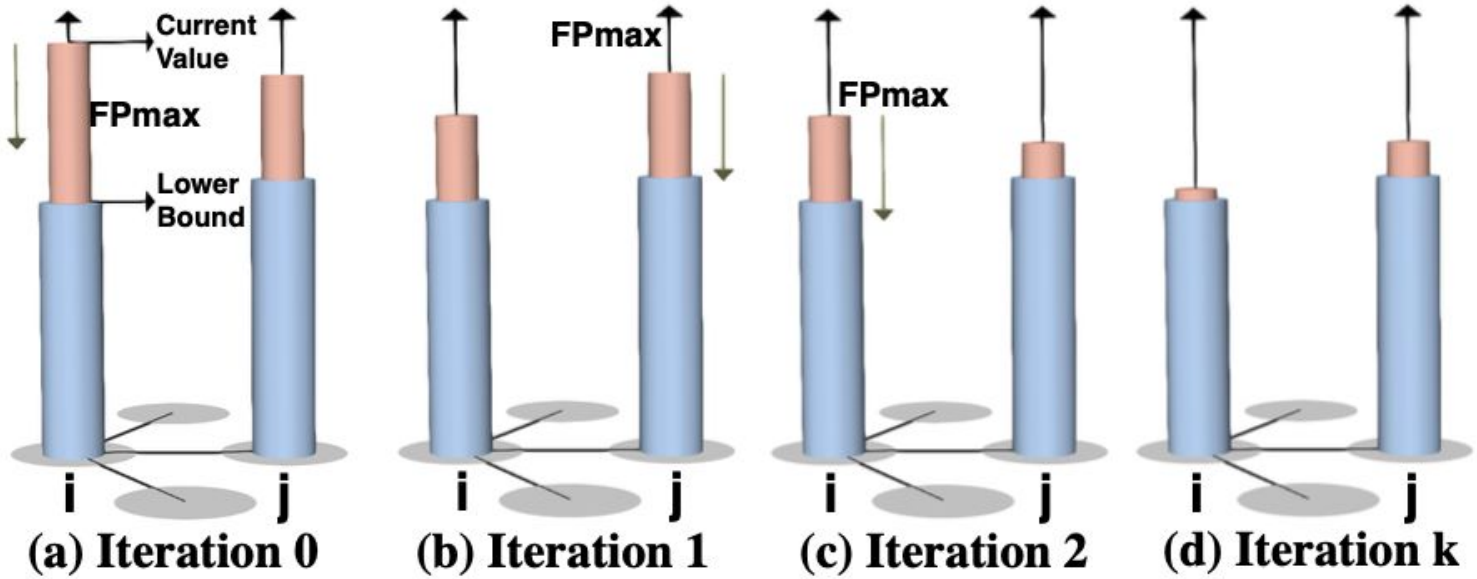
Decompression



Same compressor, same error bound ξ

After Li, Liang, Wang, Qiu, Yan and Guo, MSz, IEEE TVCG 2025 (VIS 2024)

MSz C-Loops for Fixing Critical Points

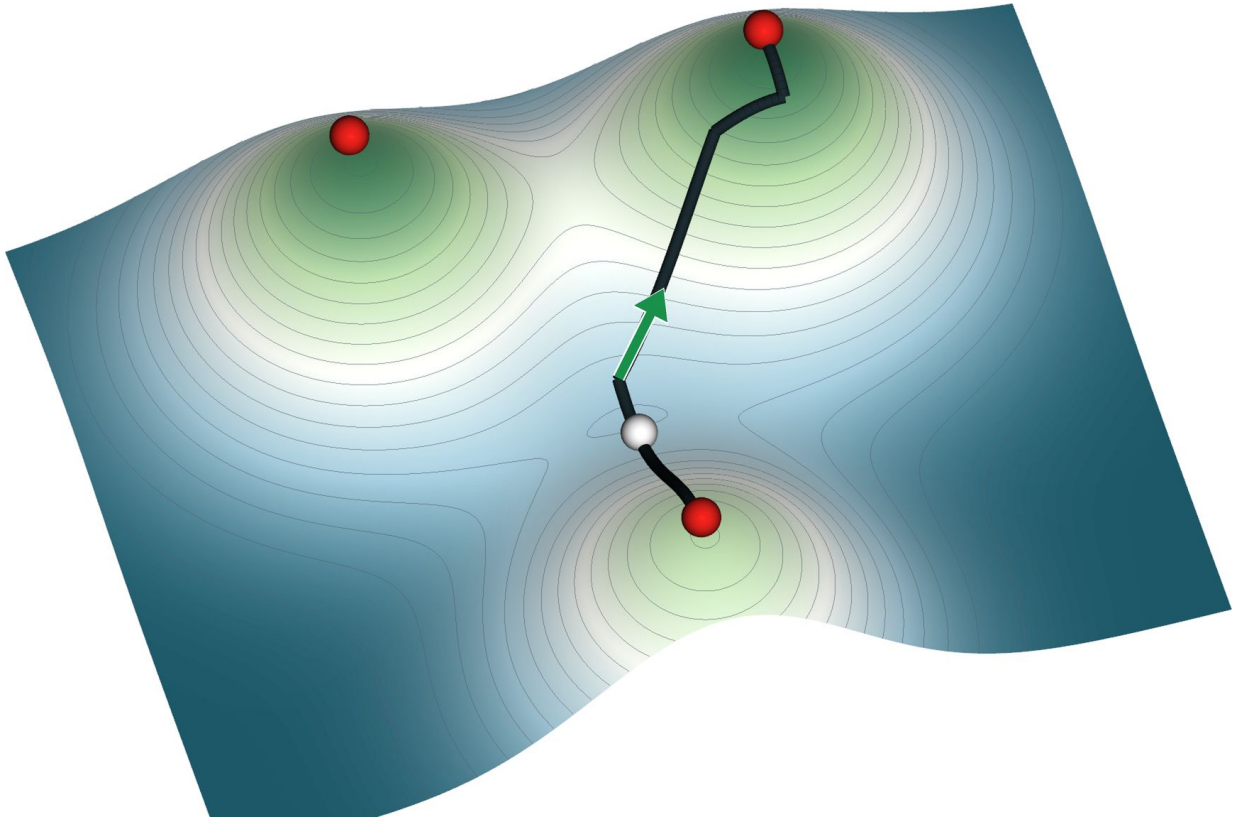


- Blue: the lower bound $f - \xi$
- Pink: the current value
- Each step lowers the offending vertex
- A vertex is settled once its neighbor's lower bound is above it
- Monotone, bounded edits guarantee convergence

All edits stay within the error bound: a value never goes below $f - \xi$.

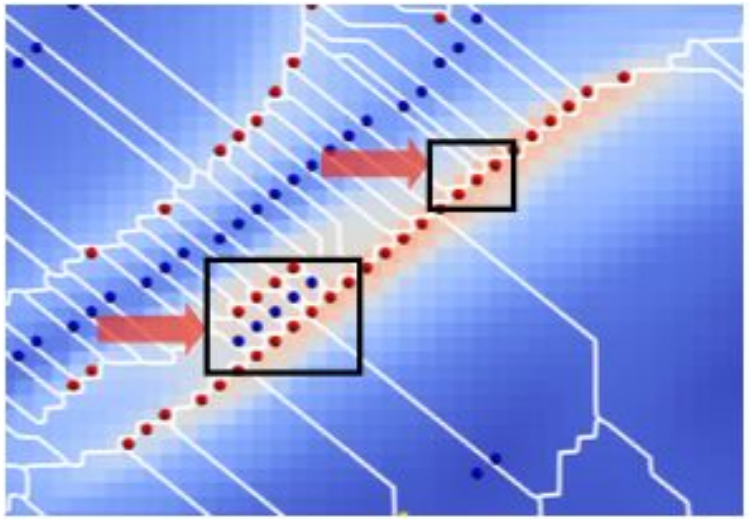
MSz R-Loops for Integral Paths

Integral path: from a point, follow the steepest ascent up to a maximum

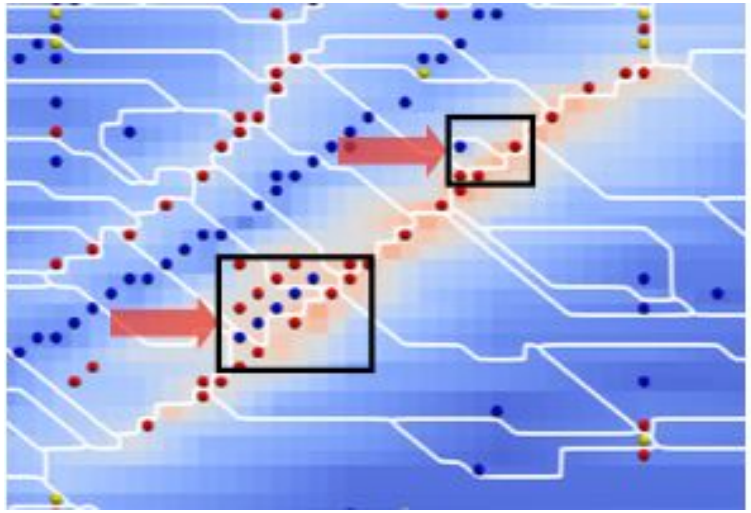


MSz: features of interest are preserved

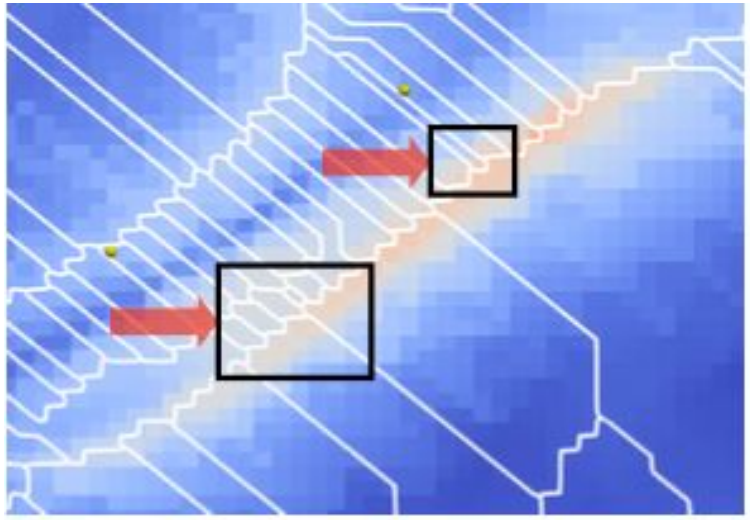
Original



SZ3



SZ3 + MSz

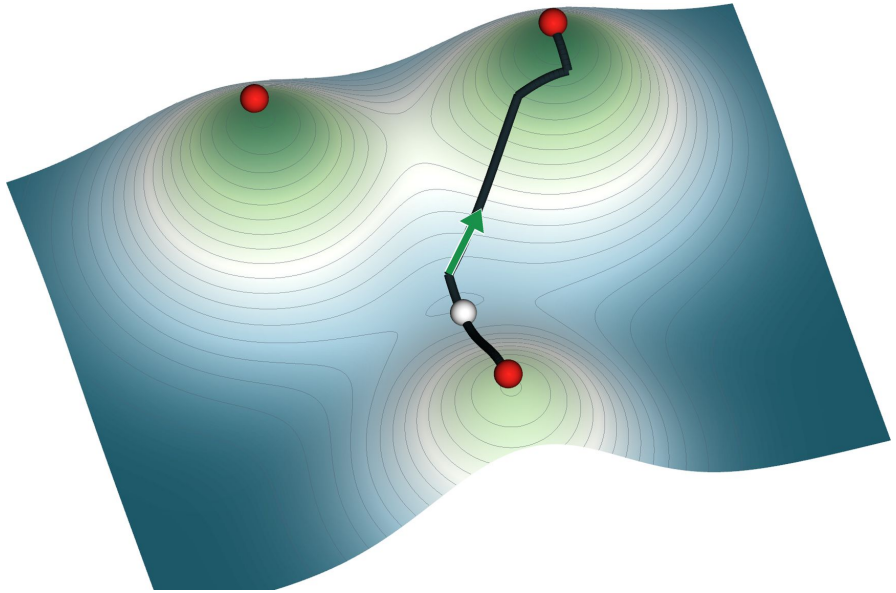


Red and blue dots: maxima and minima. SZ3 adds false extrema and moves the segment borders; **after MSz, both match the original.**

Li, Liang, Wang, Qiu, Yan and Guo, MSz, IEEE TVCG 2025. Climate data; boxes and arrows mark the differences.

pMSz: relaxing R-loops to avoids computing integral path

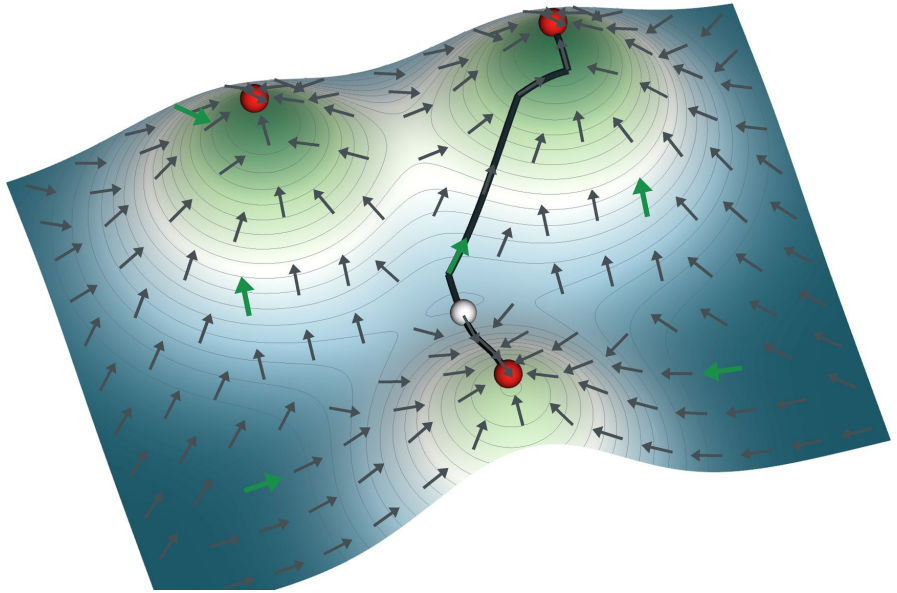
Fix trouble maker → correct path



✗ Trace each path

one path at a time → slow

② Fix all wrong directions → correct path



✓ Check every direction

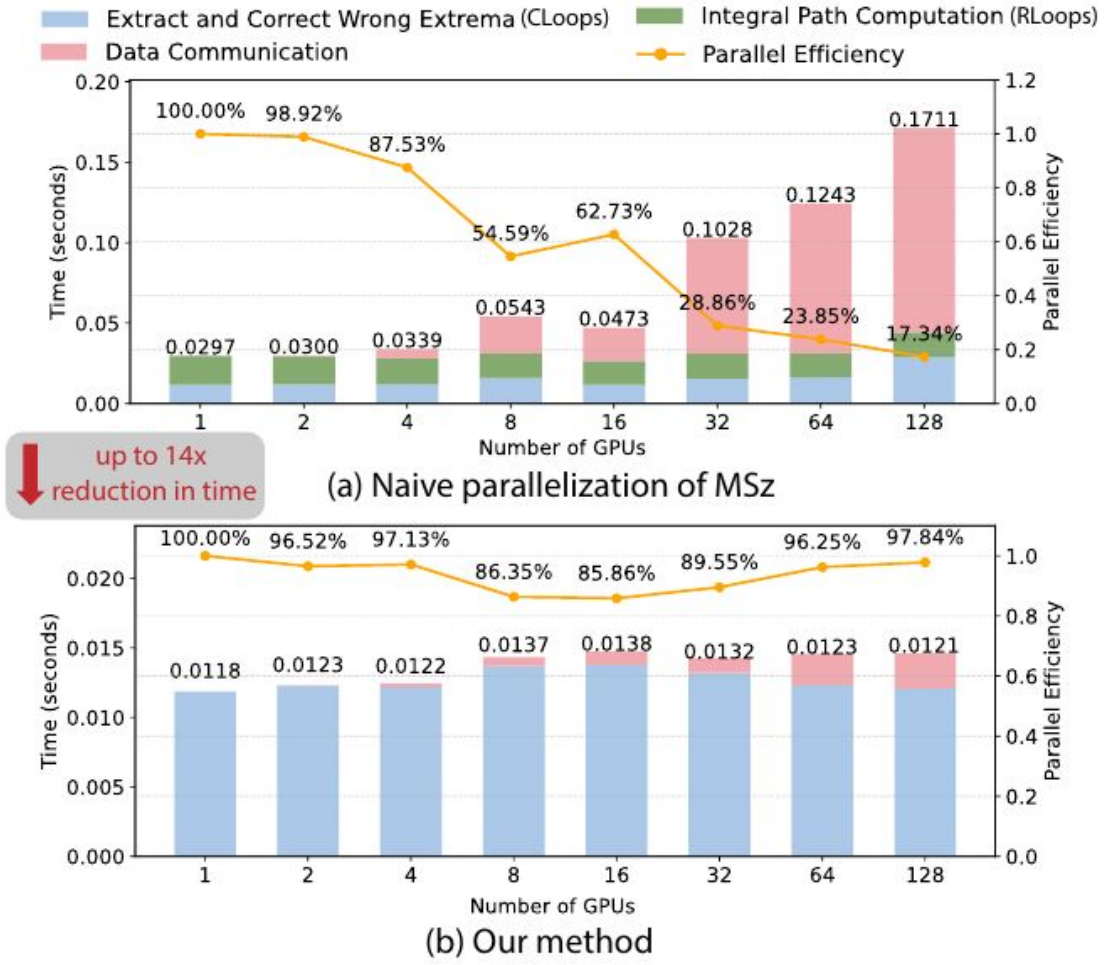
all points at once, no tracing

Key point: same directions ⇒ same paths, without tracing any path

Li et al., pMSz, IEEE IPDPS 2026

pMSz: Scale beyond a single GPU

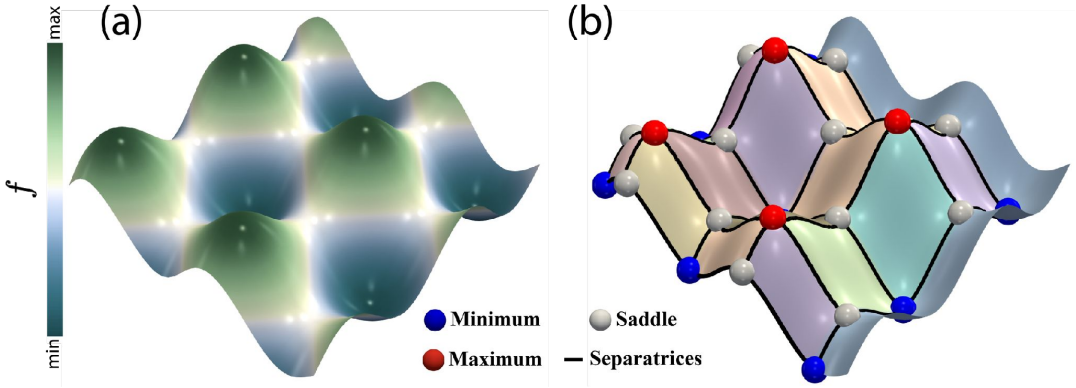
- MSz runs on one node or GPU: a few GB at most
- Naive distributed MSz: < 20% efficiency at 128 GPUs
- MSz takes up to 60% time on integral path tracing; pMSz solved the scaling issue



Li, Xia, Liang, Wang, Underwood, Di et al., pMSz, IEEE IPDPS 2026

DMTz: Preserving Morse-Smale Complexes

- Morse-Smale **segmentations** and Morse-Smale **complexes** are related tools but differ in mathematical structures
- To further preserve MS complexes, we need to ensure saddles and saddle-extrema relations



- Join this afternoon's paper session to learn more about DMTz

That's way too big!

Session chair: Peer-Timo Bremer, Grace Guo
Room: America center
Time: Thursday, Nov 12 @ 15:00 - 16:30 (US/Eastern)

15:12 - 15:24

Preserving Discrete Morse-Smale Complexes in Error-Bounded Lossy Compression

Authors: Yuxiao Li, Mingze Xia, Xin Liang, Bei Wang, Hanqi Guo

▶ **Abstract**

Concurrent Preservation of Multiple Features

Method	Contour tree	Extremum graph	GPU and distributed	Iteration bound
TopoSZ, TopoA rebuild the tree	Yes	No	No	No
MSz, pMSz edit the data	No	Partly	Yes	No
EXaCTz	Yes	Yes	Yes	Yes

GOAL

- **Preserving two topological descriptors, namely merge trees and extremum graphs concurrently**
- **Join the afternoon paper session to learn more about EXaCTz**

That's way too big!

Session chair: Peer-Timo Bremer, Grace Guo
Room: America center
Time: Thursday, Nov 12 @ 15:00 - 16:30 (US/Eastern)

15:48 - 16:00
EXaCTz: Guaranteed Extremum Graph and Contour Tree Preservation for Distributed and GPU-Parallel Lossy Compression

Best Paper

Authors: Yuxiao Li, Mingze Xia, Xin Liang, Bei Wang, Hanqi Guo

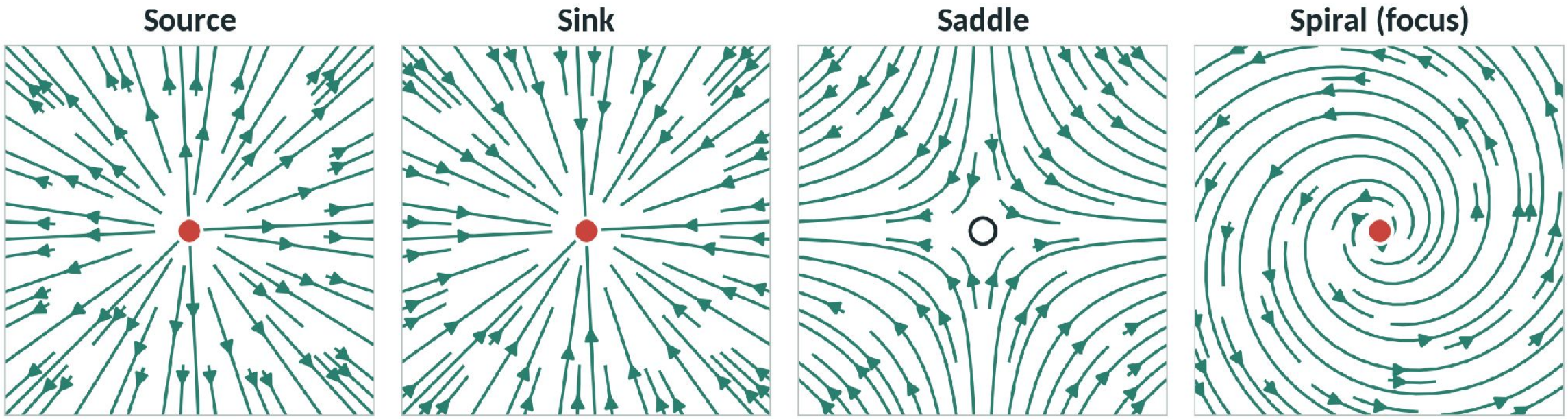
▶ **Abstract**

Vector and tensor fields

cpSZ · cpSZ-SoS · TspSZ · trajectories · TFZ

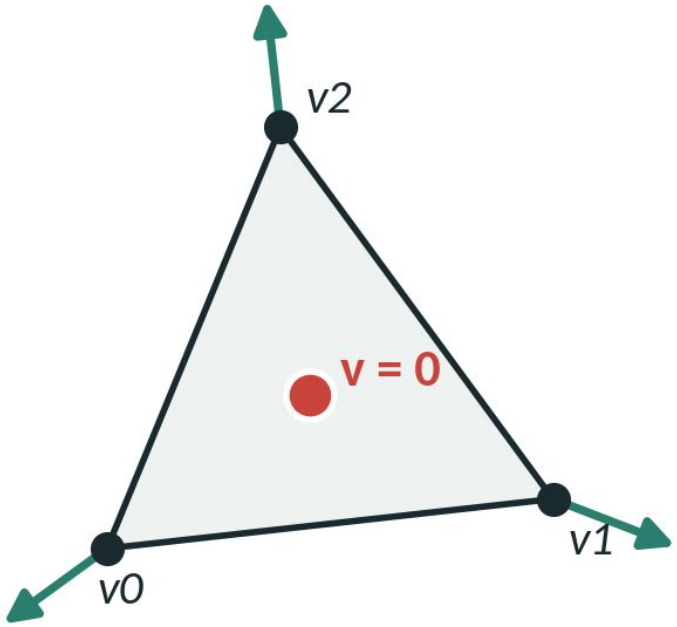
VIS2026

Critical points in vector fields



- Critical point: where v = 0; its type comes from the Jacobian
- Separatrices: streamlines that start at saddles

Critical Points in PL Vector Fields



A cell holds a critical point iff the zero vector is inside the convex hull of its vertex vectors

- Test: the signs of a few determinants
- A small error can flip one sign

The extraction of critical points involves finding zero points in each linearly interpolated cell:

$$\mathbf{V} \cdot \boldsymbol{\mu} = \begin{bmatrix} u_0 & u_1 & u_2 \\ v_0 & v_1 & v_2 \end{bmatrix} \begin{bmatrix} \mu_0 \\ \mu_1 \\ \mu_2 \end{bmatrix} = \mathbf{0} \text{ and } \mu_0 + \mu_1 + \mu_2 = 1, \quad (2)$$

Preserving critical points = preserving the signs of determinants.

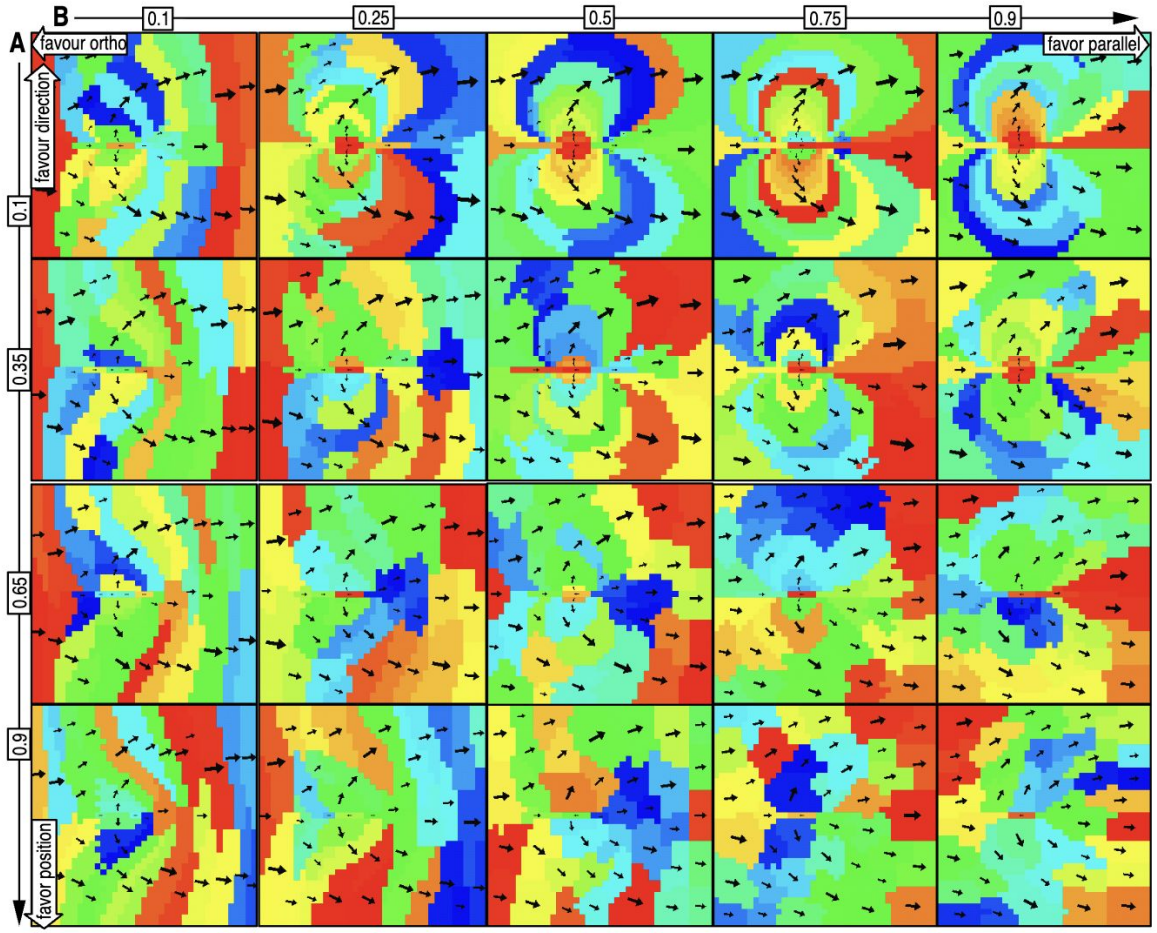
Simplified Representation of Vector Fields

```
ClusterSet s;
for (all cells cell_i in dataset)
{
    c = makeCluster(cell_i);
    set level of c to 0;
    add c to s;
}

for (all clusters c_i in s)
  for (all clusters c_j neighbours of c_i)
  {
     e = clustering_error(c_i,c_j);
     insert pair (c_i,c_j) in increasing order
     of error e in a hash-table;
     mark c_i and c_j as NOT_CLUSTERED;
  }

int l=0;
for (all pairs (c_i,c_j) in increasing order of
     error in the hash-table)
  if (both c_i and c_j are NOT_CLUSTERED)
  {
     c = mergeClusters(c_i,c_j);
     set level of c to l++;
     mark c_i and c_j as CLUSTERED;
     for (all neighbours n_i of c)
     {
       e = clustering_error(c,n_i)
       insert pair (c,n_i) in order in hash-table;
     }
  }

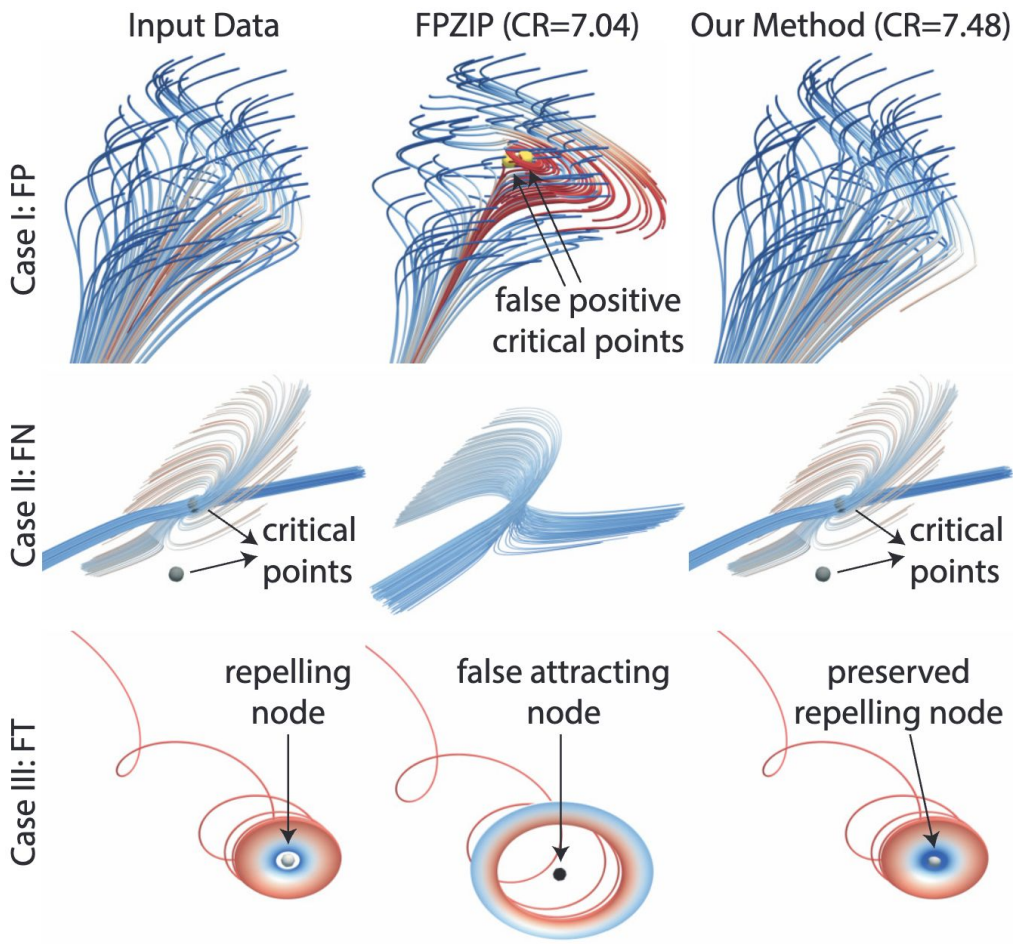
return c as root of tree;
```



Telea and van Wijk,

cpSZ: critical points preserved by construction

- Critical point preserved: Every critical point stays in its cell and keeps its type
- No guarantees on topological skeleton in cpSZ

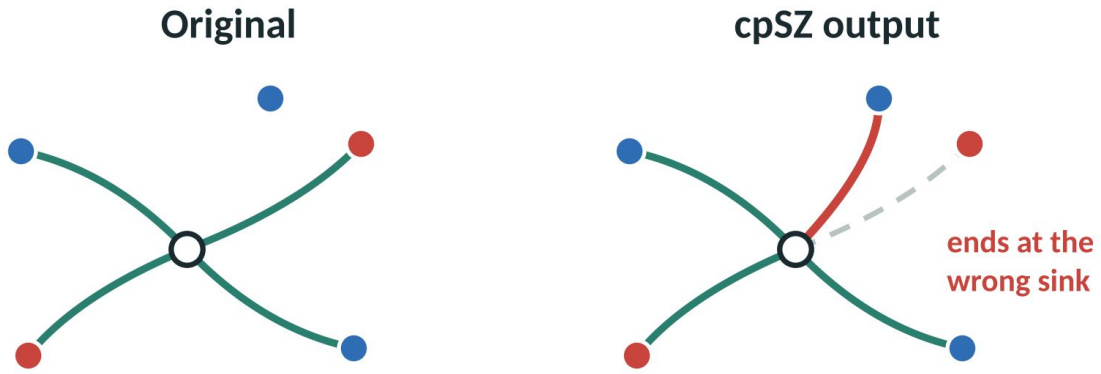


Liang, Guo, Di, Cappello et al., Toward feature-preserving 2D and 3D vector field compression, IEEE PacificVis 2020; extended in IEEE TVCG 2023

TspSZ: the whole topological skeleton

Critical points alone are not the topology.
Separatrices matter too.

- cpSZ keeps the critical points, but separatrices drift
- TspSZ keeps the separatrices too



White: saddle. Blue: sinks. Red: sources.

After Xia, Jiao, Wang, Liang, Li and Guo, TspSZ, IEEE ICDE 2025

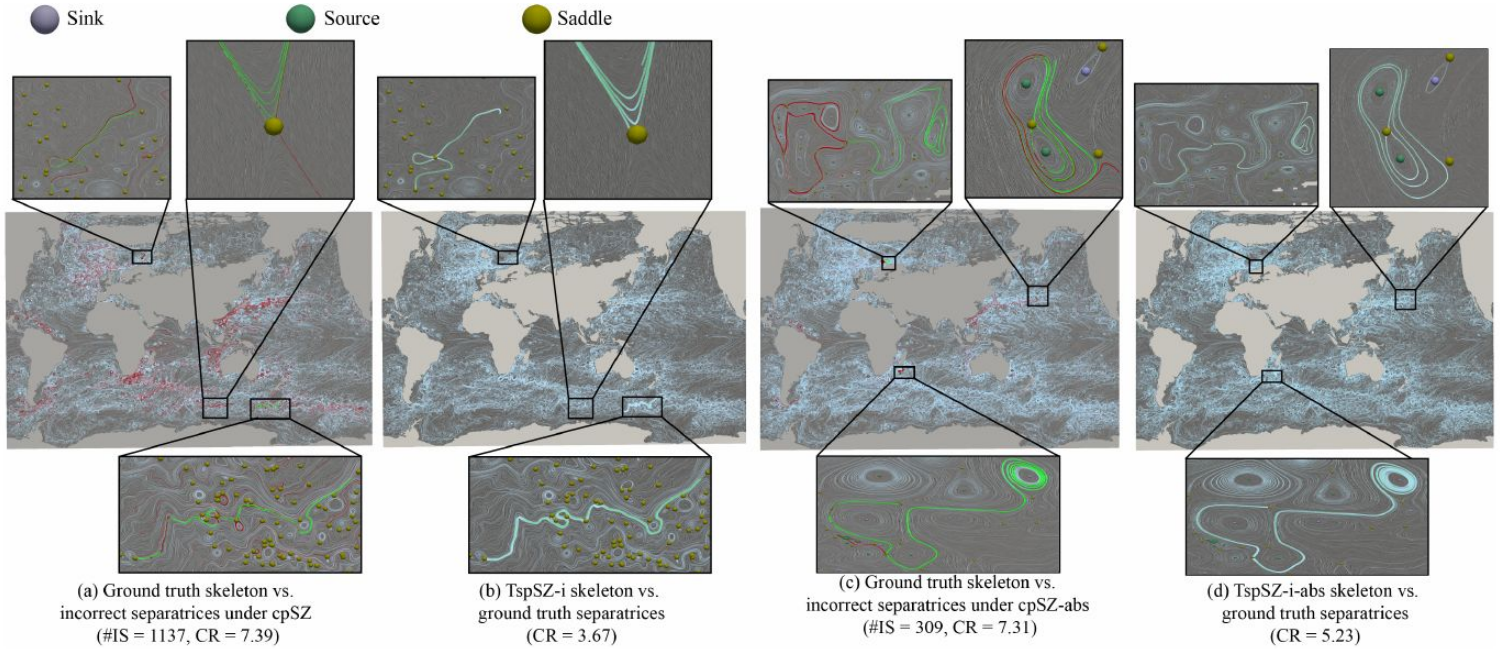
TspSZ: results

7.7×

compression ratio, skeleton preserved

0

wrong separatrices (cpSZ: 912 and 14,576)



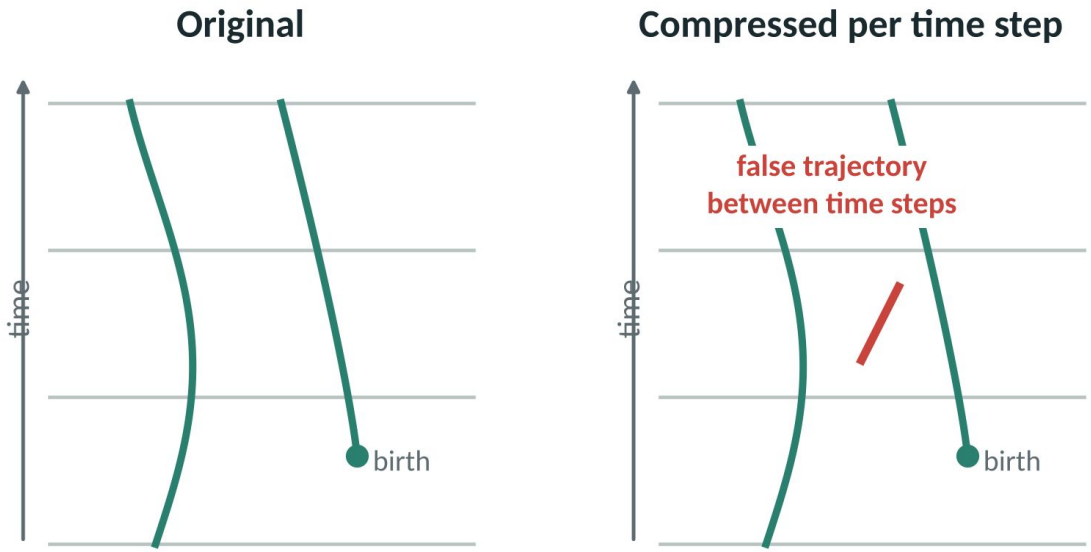
Absolute error control; OpenMP parallel.

Figure: Xia et al., IEEE ICDE 2025 (ocean data; red: incorrect separatrices, green: ground truth)

Time-varying fields: preserving critical point trajectories

Critical points move, appear, merge and vanish over time

- Trajectories track cyclones, eddies and vortices
- Checking each time step is not enough: we check space-time cells



After Xia, Wang, Li, Liang, Jiao and Guo, arXiv:2510.25143

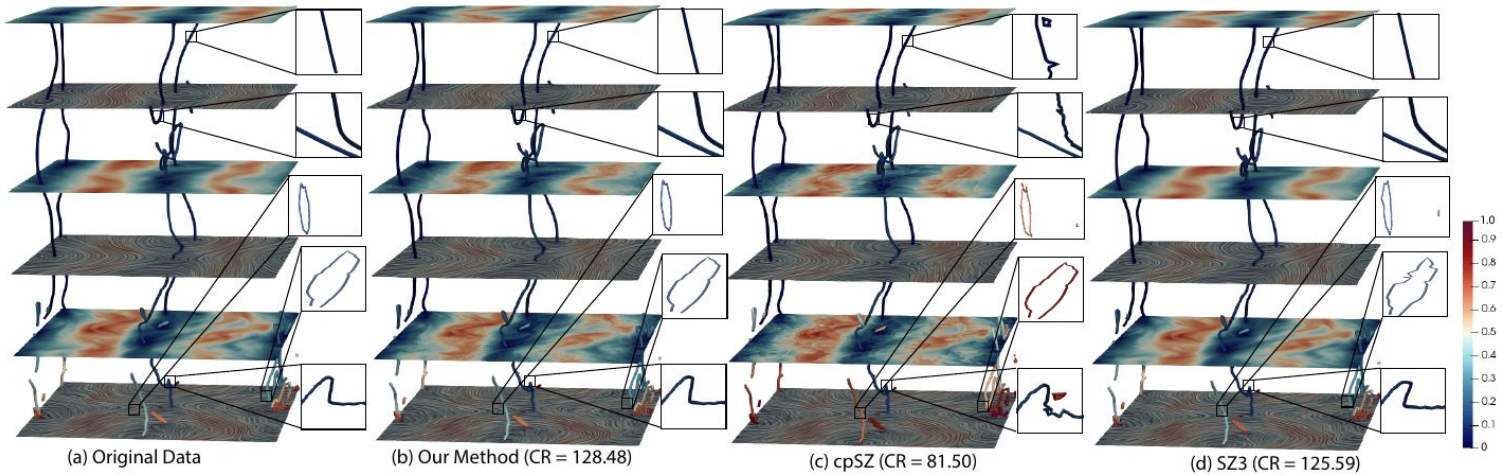
Trajectories: results

124.48×

compression ratio, all trajectories kept

56×

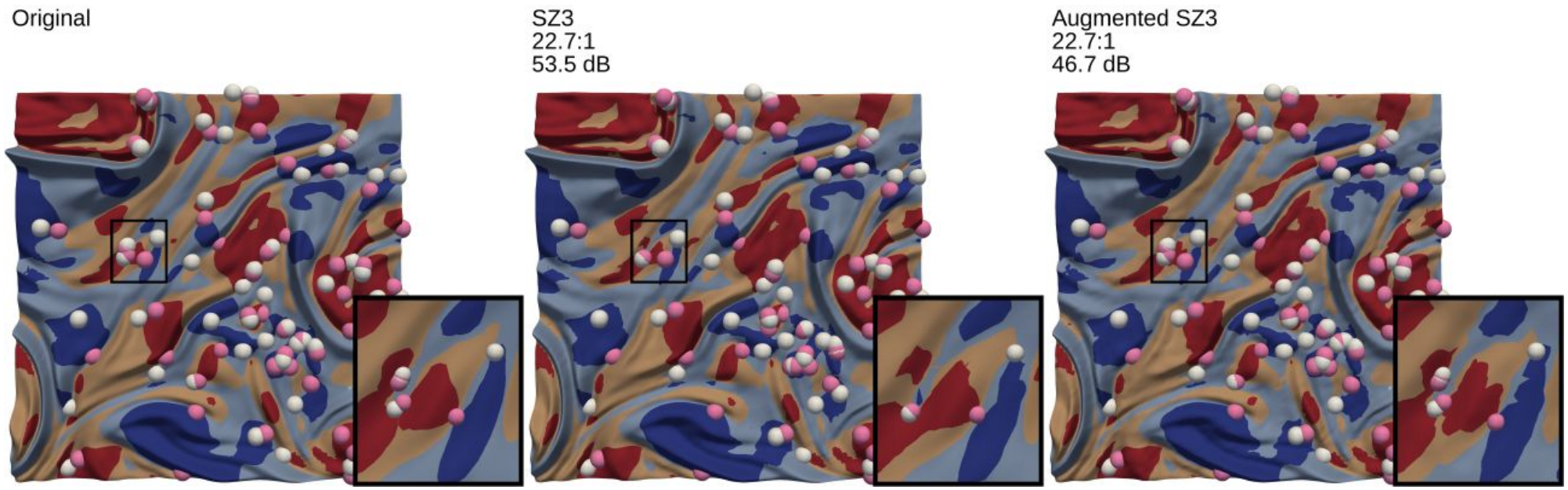
higher than the best lossless compressor



Semi-Lagrangian predictor: up to 4.31× over Lorenzo.

Figure: Xia, Wang, Li, Liang, Jiao and Guo, arXiv:2510.25143. Zero false cases in time steps and time slabs on five datasets.

Beyond vectors: TFZ for 2D tensor fields

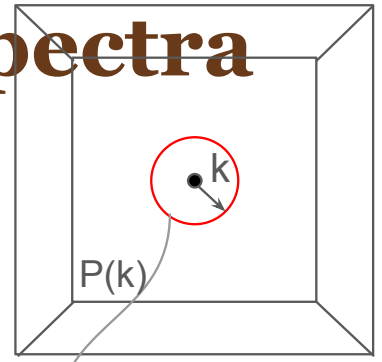


- 2D symmetric and asymmetric second-order tensor fields on triangular meshes
- Symmetric fields: preserves the degenerate points. Asymmetric fields: preserves the eigenvector and eigenvalue graphs
- Works cell by cell, preserving local topology, which gives global guarantees; augments SZ3 and SPERR

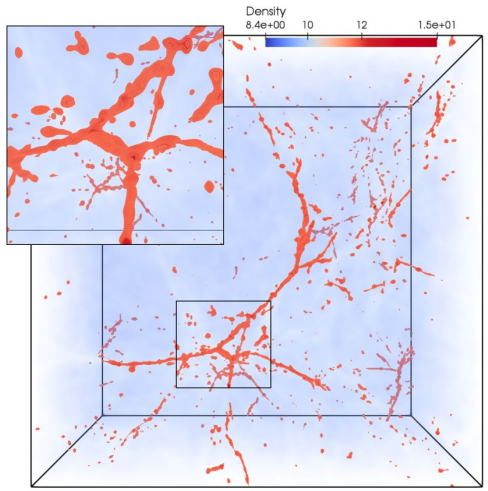
Figure: Gorski, Liang, Guo and Wang, TFZ, IEEE TVCG 2026 (VIS 2025). Left to right: input, SZ3, augmented SZ3.

Beyond Topology: FFT and Power Spectra

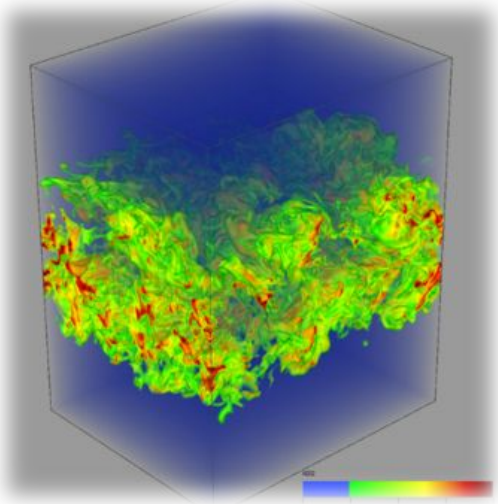
- Similar to topological features, bounded error would lead to significant error in frequency domains



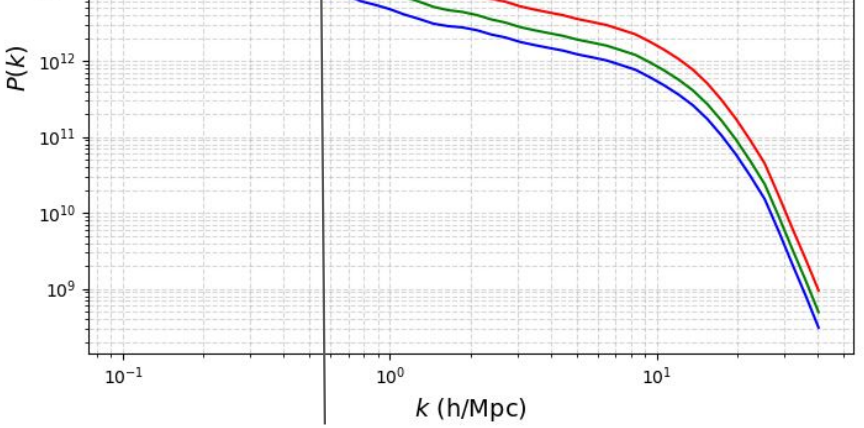
frequency domain



Cosmology (Nyx)



Combustion (S3D)



Method for FFT Correction Schemes

- spatial error
$$\varepsilon_n = \hat{x}_n - x_n$$
- frequency error
$$\delta_k = \hat{X}_k - X_k$$

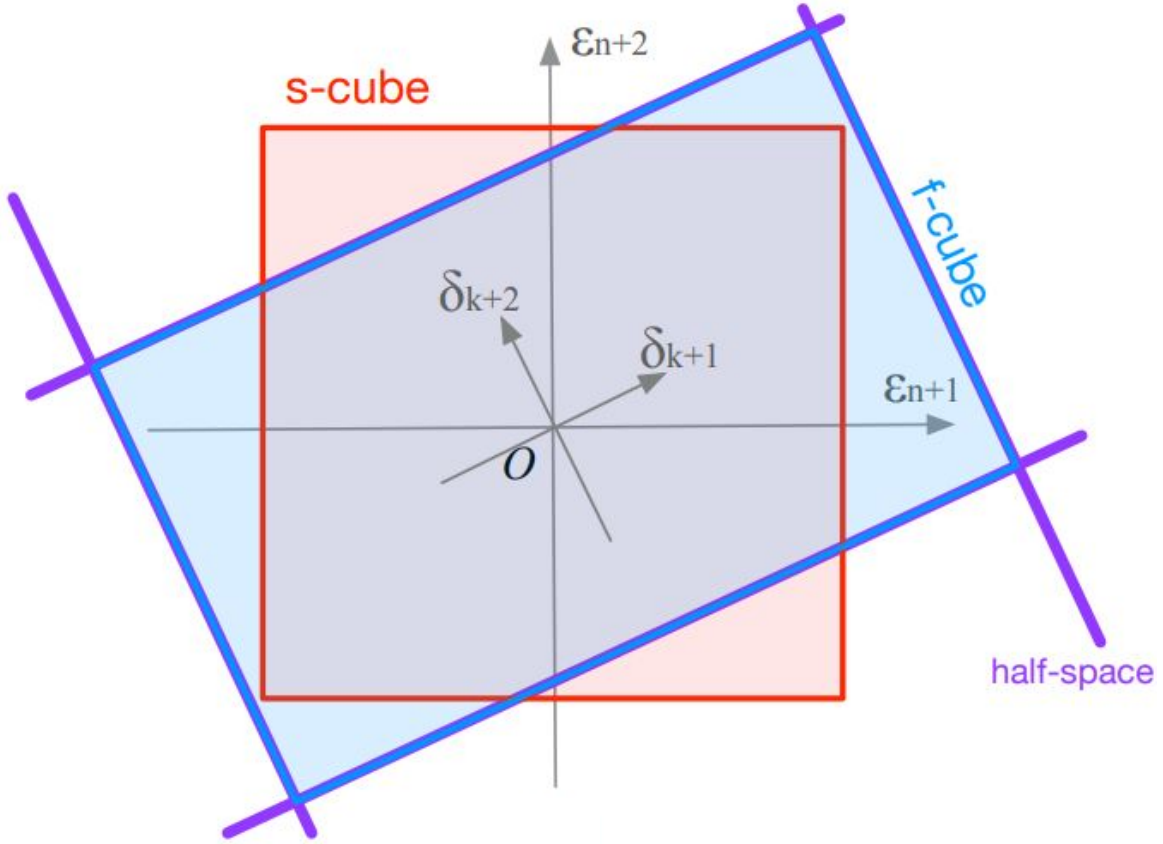
- spatial constraints
$$|\hat{x}_n - x_n| \le E$$
- frequency constraints
$$|\hat{X}_k - X_k| \le \Delta$$

s-cube constraints

$$-E \le \epsilon_n \le E \text{ for } n \in \{0, 1, \cdots, N-1\},$$

$$\begin{cases} -\Delta \le \sum_{n=0}^{N-1} \cos(\frac{2\pi k}{N} n) \cdot \epsilon_n \le \Delta \text{ for } k \in \{0, 1, \cdots, N-1\}, \\ -\Delta \le \sum_{n=0}^{N-1} \sin(\frac{2\pi k}{N} n) \cdot \epsilon_n \le \Delta \text{ for } k \in \{0, 1, \cdots, N-1\}, \end{cases}$$

f-cube constraints

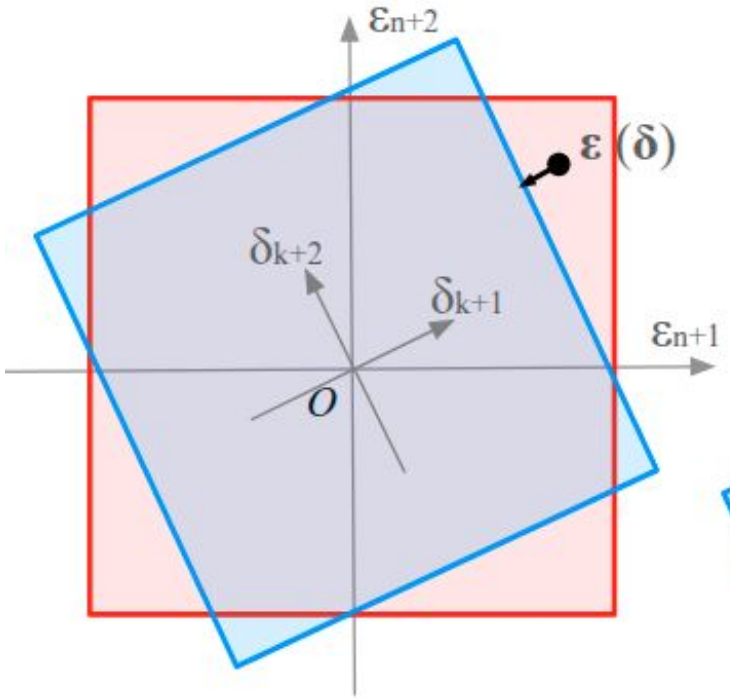


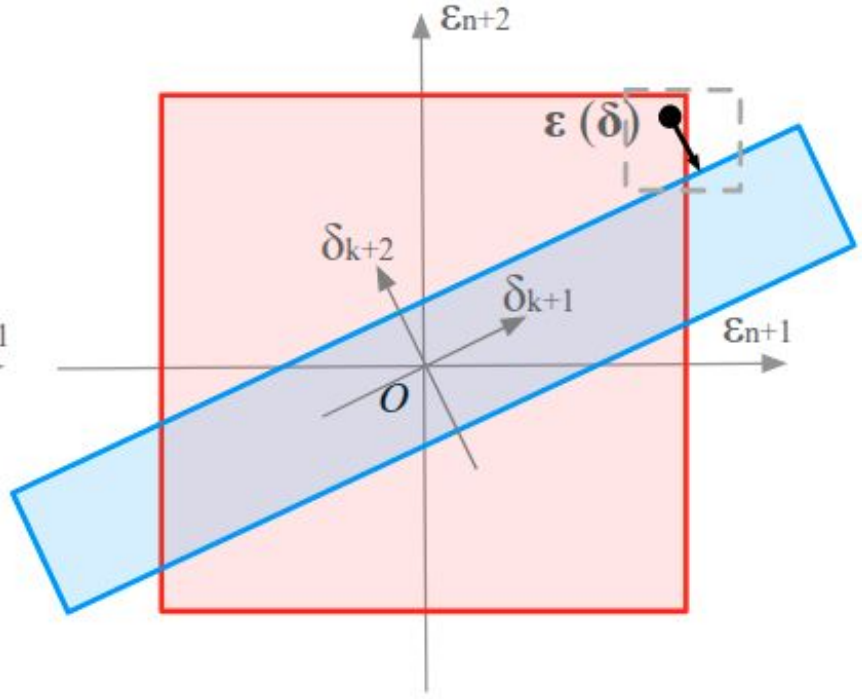
axes: spatial errors ε_n's

Iterative Projection for FFT Correction

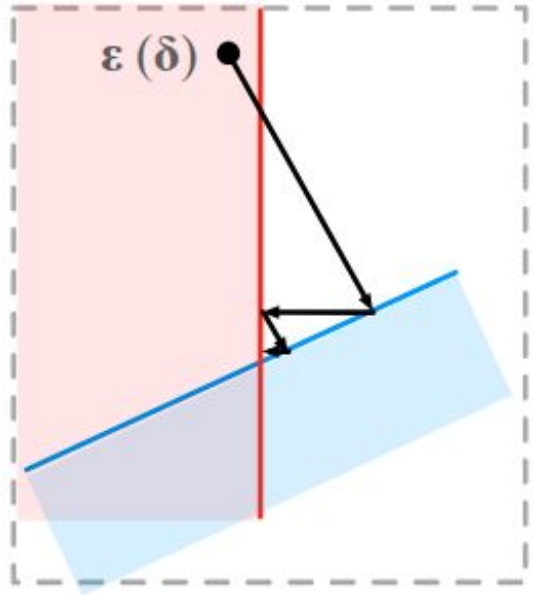
Alternating projection

Case 1





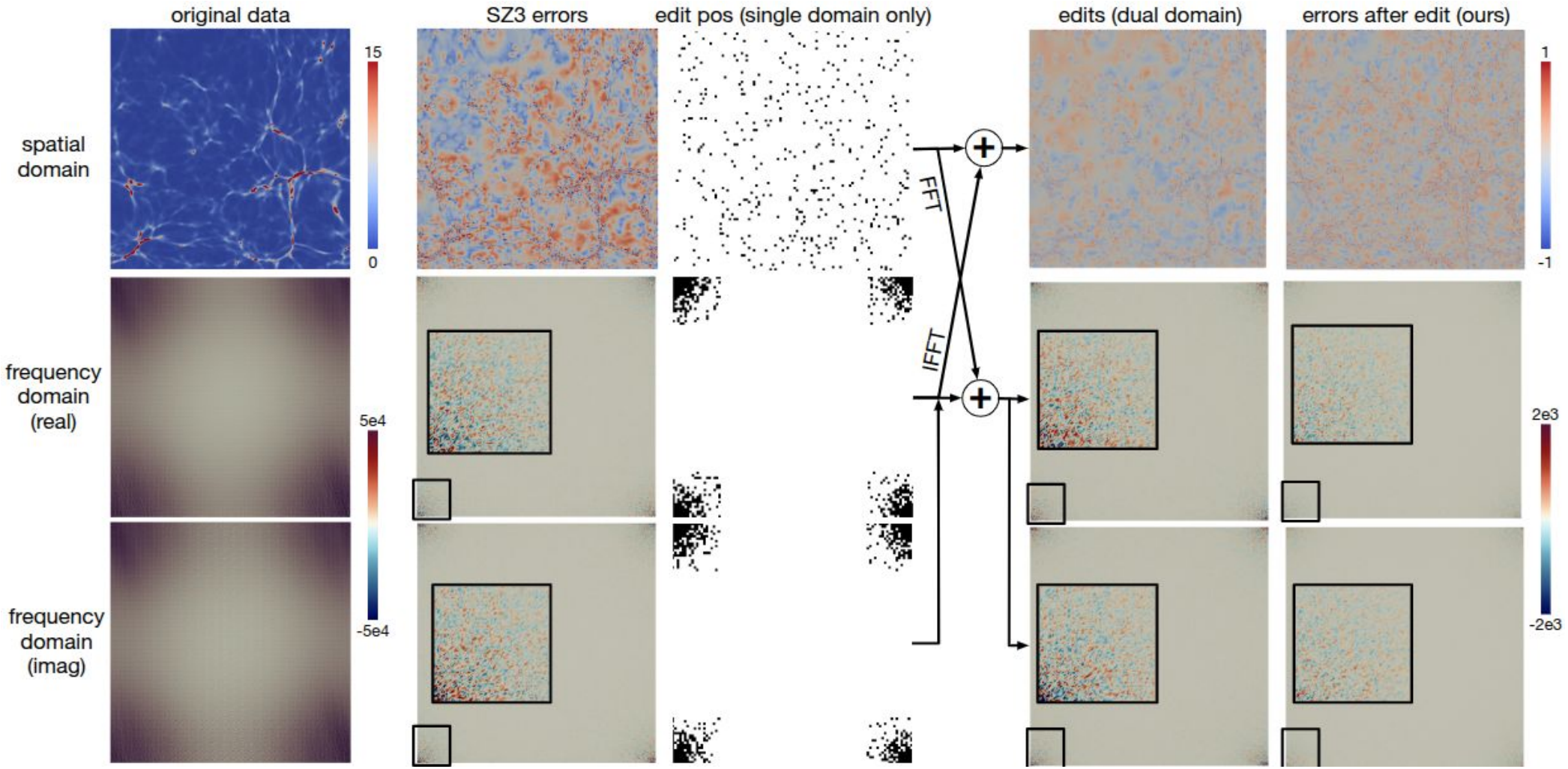
Case 2



{ spatial correction terms (along ε_n)

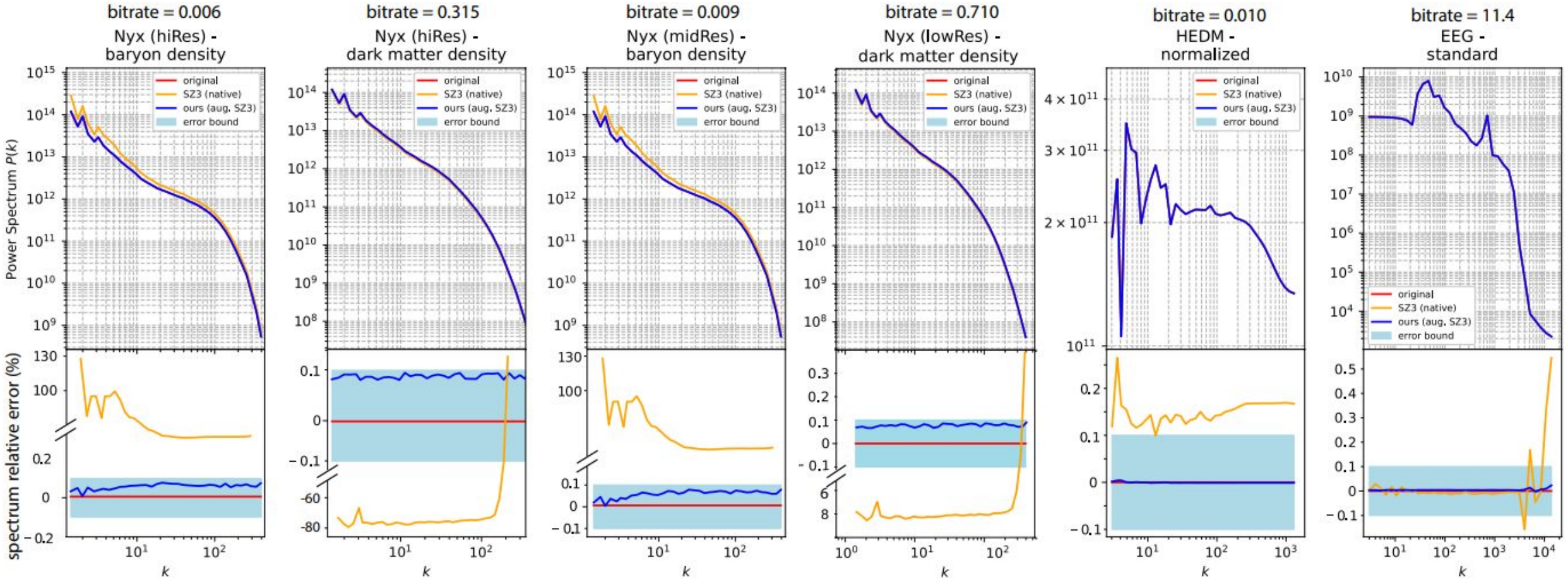
frequency correction terms (along δ_k)

Corrections in both Spatial and Frequency Domain



Evaluation - Power Spectrum

Better preservation of power spectra under the same bitrate



Ren et al., “FFCz: Fast Fourier Correction for Spectrum-Preserving Lossy Compression of Scientific Data.” In IPDPS '26: Proceedings of the 40th IEEE International Parallel and Distributed Processing Symposium, pages 130-143, 2026.

Summary

VIS2026

Which method preserves what

Method	Data	Preserves	Approach	Parallelization
TopoQZ	Scalar	Critical point pairs above a persistence threshold	Constrain	–
TopoSZ	Scalar	Contour tree (after simplification)	Constrain	Serial
TopoA	Scalar	Contour tree	Correct	Serial
MSz, pMSz	Scalar	Morse–Smale segmentation	Edit	GPU; distributed
DMTz	Scalar	Morse–Smale complex, in tiers	Edit	GPU
EXaCTz	Scalar	Extremum graph and contour tree	Edit	GPU; distributed
cpSZ, cpSZ-SoS	Vector	Critical points	Constrain	Distributed
TspSZ	Vector	Critical points and separatrices	Constrain + correct	Shared memory
Trajectories	Vector, time	Critical point trajectories	Constrain	Serial

Takeaways and open problems

Takeaways

- A pointwise error bound does not protect topology
- Topology is decided by orderings and signs, so it can be protected by constraining or correcting them
- Choose the descriptor your analysis needs; pay only for that
- The cost is compression ratio and time, and both are now practical

Open problems

- Memory footprint and in situ use
- Tighter cost prediction before compressing
- More data types: time-varying scalar fields, 3D tensors, unstructured meshes
- Preserving several descriptors at once